

Cryptocurrency Volatility: A Review, Synthesis, and Research Agenda

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Cryptocurrency Volatility: A Review, Synthesis, and Research Agenda

Abstract

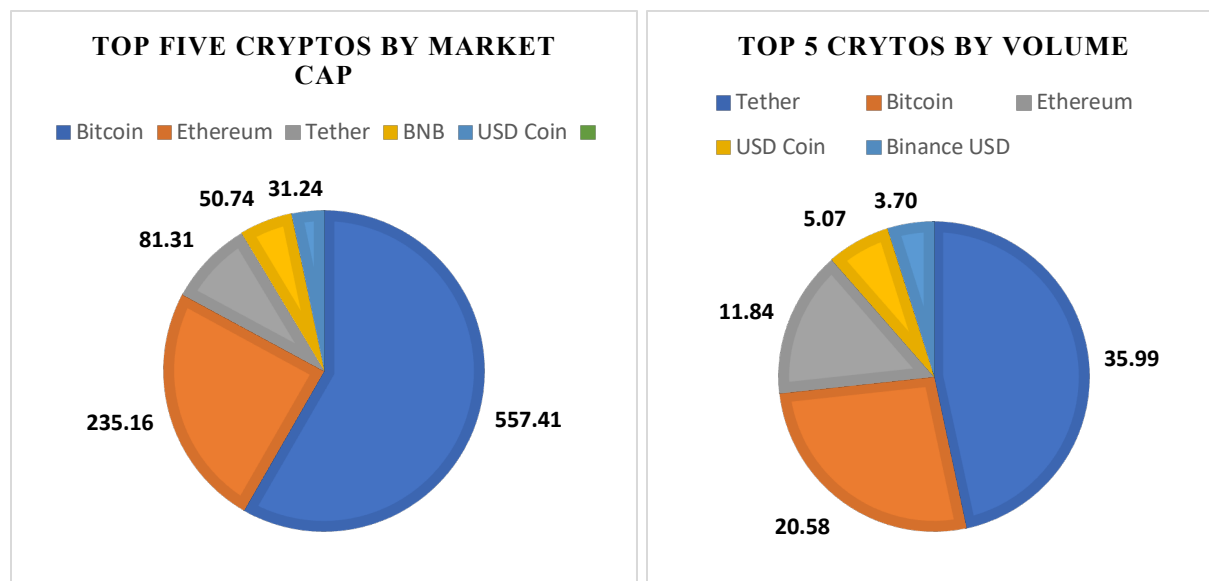
This paper takes part in the ongoing debate on the newly emerging field of financial technology by systematically reviewing 164 articles on cryptocurrency volatility during the period from 2016 to December 2022. This paper also aims to enlighten academics and practitioners about the beneficial insights gained from cryptocurrency volatility research, identify existing research gaps, and propose a new research agenda in the subject of study. To this end, realized volatility, almost all stylized facts, implied volatility, stochastic volatility, and drivers of volatility are discussed. Finally, we propose that future researchers concentrate on high-frequency data (i.e., hourly, minutely, and secondly), the use of machine learning models, crypto derivatives, crypto individual investor behavior, the impact of the new existence of institutional investors, stablecoins, and the evaluation of the forecasts of cryptocurrency volatility.

Keywords: Cryptocurrencies; Bitcoin; Volatility; Financial risk; Review paper; Research agenda

JEL classification: G11, G14, G15, G17, G18, G28

1- Introduction

Nakamoto (2008) proposed the concept of Bitcoin in 2008. Since then, cryptocurrencies in general and Bitcoin in particular have gained unprecedented attention from computer experts, academicians, economists, entrepreneurs, and investors because of their unique features. These features are an attractive combination of innovative information technology and cybersecurity infrastructure, as well as investability as a new asset. Nonetheless, the unregulated and decentralized nature of cryptomarkets provides them with an extraordinary level of uncertainty and imposes serious challenges on not only conventional financial institutions but also the whole global financial system and well-established monetary policies (Urquhart, 2016; Bariviera, 2017).



Top 5 cryptos by market cap (numbers in billions)

Top 5 cryptos by dollar volume (numbers in billions)

Figure 1. Top 5 cryptos by market cap and by volume as of 20/04/2023.

Source: authors

Volatility is a central issue in finance. As a result, estimating and forecasting volatility is critical for managing risk, asset pricing, investment strategies, market efficiency, and selecting portfolios. The substantial increase in traders, other market participants, media coverage, the number of new crypto currencies, and the volume and value of transactions over the last few years has maximized the motivation for researching cryptomarket volatility. In this vein, despite several manipulations and frauds being detected, an ongoing academic and practical debate has arisen concerning the evolution of cryptomarkets, why investors trade these products, the legitimacy of trading, and the monetary, ethical, and technological foundations of these markets (Poon &

Granger, 2003). Thus, advancing knowledge and improving our understanding of cryptomarket volatility and how this new financial world operates are of particular importance. Crypto investors, traders, and other market participants take volatility into account in asset pricing and developing investment strategies, and there are some beneficial policy implications. In light of the growing number of papers in the literature, we believe that it is critical and well-timed to review the existing body of research. Despite this growth in scientific research on cryptocurrency volatility, we argue that the literature on cryptocurrencies is still in its infancy stage and is characterized by a deficient, disconnected, disparate, and divided body of knowledge. The constant surge of literature on cryptocurrency volatility has made the systematic review of this subject of study troublesome and increased the need for a comprehensive overview of this stream of research.

The starting point of this paper is to define and review realized volatility, including the datasets used in the literature, the known centralized and decentralized crypto exchanges, idiosyncratic risk, and the gaps in the existing literature. Next, we pay attention to a wide range of stylized facts, namely, volatility clustering, volatility persistence, asymmetric volatility, leverage effect, volatility spillover, mean reversion, structural breaks, extreme volatility, and tail dependence. For each stylized fact, we present a straightforward definition, the main streams in the literature, and the current research gaps that relate to the concerned stylized fact. Similarly, implied volatility, stochastic volatility, and the key drivers of volatility are discussed in terms of the corresponding definitions for each, the relevant streams of the existing literature, and some of the gaps. Then, we shed light on the implications of this research in the discussion section, with a focus on the unique stylized facts of cryptomarkets, the econometric models used, market efficiency, and hedging and safe haven capabilities. After that, we propose several future research directions related to volatility in cryptomarkets in the research agenda section, including replicability, reproducibility, and robustness studies, high-frequency data research, advanced forecasting techniques, cryptomarket derivatives, and investor behavior. Finally, we summarize the concluding remarks in the final section.

The purpose of this paper is threefold. First, it provides a comprehensive and systematic review of all kinds of cryptomarket volatility (e.g., asymmetric volatility, volatility clustering, realized volatility, etc.) from 2016 to December 2022, including the seminal contributions made, datasets, econometric models, key findings, and implications, in order to present the inconsistencies and gaps in the current body of knowledge, synthesize different findings, and provide readers with a clear picture of the current state-of-the-art. We make 2016 our starting point

because getting papers before that was hard, and we were able to get three papers only in 2016 (see figure 2). Specifically, this paper aims to provide an answer to the following questions: What do we know about cryptocurrency volatility? What questions do we have to address to advance our knowledge of this newly emergent subject of study? Secondly, it sheds some light on the development of academic research on cryptomarket volatility, which helps to improve our understanding of the current body of knowledge. Thirdly, based on the previous two purposes, it further aims to identify enormous research gaps and several research directions to further motivate the theoretical development of cryptomarket volatility.

The methodology for the selection of papers is taken from Brauer (2006), Haleblan, Devers, McNamar, Carpenter, and Davison (2009), and Schweizer and Nienhaus (2017). This methodology contains three steps. First, the period from 2016 to December 2022 is considered the review period to guarantee the insertion of all papers published during this period. Second, we select the kinds of cryptocurrency volatility (i.e., volatility clustering, volatility spillover, structural breaks, etc.) on which we focus our search because they represent the key mainstreams of volatility research. Third, we use the following keywords in our search: cryptocurrency volatility, bitcoin volatility, cryptocurrency financial risk, price fluctuations of cryptocurrencies, and cryptocurrency risk. These keywords are employed to collect research articles and working papers from Google Scholar, Elsevier, and SSRN. All papers are considered for review, except those whose journals did not take part in the ABS journal ranking, because we define these journals as poor-quality journals and those that are not empirical. Fourth, we employ Scopus instead of Google Scholar for bibliometric analysis (section 2) because Google Scholar is afflicted by quality control issues. In the same vein, citations reported by Scopus are strongly associated with library judgements and information science scholars than those reported by Google Scholar (Baker, Kumar, & Pandey, 2020). This amounts to 164 articles, including 158 published journal articles and six working papers. We limit our search to English papers. To make sure that cryptocurrency volatility is the key topic of research, we check each paper in terms of the contribution made, title, abstract, and keywords. A practical screen followed to remove book reviews and literature reviews and to focus on papers that obviously addressed the volatility of cryptocurrencies. After that, the search was restricted to papers containing field of research. Finally, we use the classification method of Poston & Grabski (2000) that classify articles based the underlying theory, research method and research topic. This results in a segmentation of the stylized facts of volatility and, in turn, the collected papers into volatility clustering, volatility persistence, asymmetric volatility, leverage effect,

volatility spillover, mean reversion, structural breaks, extreme volatility, and tail dependence. All other papers are distributed separately for realized volatility, implied volatility, stochastic volatility, and drivers of volatility.

This review makes many contributions to the prior literature. First, some reviews addressed the cryptocurrency subject of study, but none of them made the price volatility of cryptocurrencies the focal point of the review. For instance, both Jalal, Alon, and Paltrinieri (2021) and García-Corral et al. (2022) conducted bibliometric analyses. The former focused on the determinants of cryptocurrency returns, cryptomarket efficiency, diversification, and regulation, while the latter focused on the cryptocurrency concept in terms of the growth of the scientific production of this new financial product, with particular attention to Bitcoin and Ethereum. Kyriazis, Papadamou, and Corbet (2020) reviewed price bubbles in the cryptomarkets. Hairudin et al. (2022) surveyed the most controversial issues of cryptocurrency research, such as acceptance, design, valuation, and legitimacy. Rejeb, Rejeb, and Keogh (2021) surveyed the role of cryptocurrencies in the modern business environment and their challenges for current financial systems. Ul Haq et al. (2021) concentrated on cryptocurrencies as a risk management tool against economic policy uncertainty (EPU) in their review. Bariviera and Merediz-Solà (2021) conducted a comprehensive bibliometric analysis of the economics of cryptocurrencies. Hossain (2021) conducted a meta-analysis of the stylized facts of cryptocurrency research from 2016 to 2020 in terms of the unique features of blockchain technology, low-cost platforms, decentralized methods of payment, and unregulated markets. Therefore, this review is the first to evaluate, integrate and organize the available knowledge on cryptocurrency volatility in general and the stylized facts of volatility in particular. Second, this paper is the first to conduct a bibliometric analysis of cryptocurrency volatility. Additionally, it is the first, to the best of our knowledge, to identify inconsistencies in the prior relevant literature and to propose many research gaps for each stylized fact, as well as several future research directions that motivate the theoretical development in the subject of study.

Based on our review, we find some research trends lack sufficient attention from the literature. Thus, researching these areas may enrich and advance the volatility theory of cryptocurrencies. This includes robustness studies, high-frequency data, machine learning models, cryptocurrency derivatives, crypto investor behavior, stablecoins, and green cryptocurrencies.

The rest of the paper is structured as follows: Section 2 presents the bibliometric analysis. Section 3 reviews the relevant prior literature on realized volatility, stylized facts, implied

volatility, stochastic volatility, and drivers of volatility. Section 4 discusses the key implications of this review. Section 5 suggests the research agenda, including many promising future research directions. Section 6 presents the concluding remarks.

2- Bibliometric Analysis (Descriptive statistics)

The bibliometric analysis of the cryptocurrency volatility collection demonstrates that escalating attention is being paid to cryptocurrency volatility over time. Figure 1 exhibits a growth in the number of research papers in the last four years compared to the first two years, with the largest number of papers displayed in 2020.

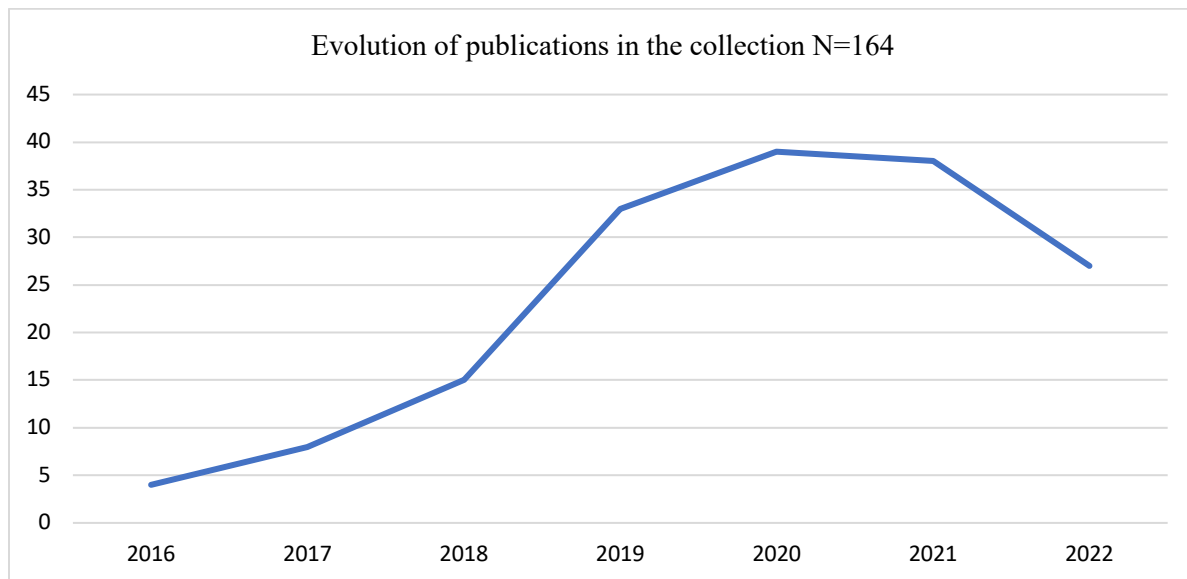


Figure 2. The cumulative number of sample publications between 2016 and 2022

Source: authors

Table 1 exhibits the top ten contributing journals in the collection of this review. It is noticed that Finance Research Letters is the largest research contributor in this review, with more than 29% of the total research production, followed by Research in International Business and Finance, Economics Letters, and International Review of Financial Analysis. The Journal of Empirical Finance and Studies in Economics and Finance are the two lowest contributors among the top journals in the collection, with three articles for each. Table 2 exhibits the total number of citations for each of the top ten articles in the collection. The seminal paper authored by Satoshi Nakamoto (2008) is by far the most cited paper, with more than 25,000 citations, followed by Dyhrberg (2016), with up to 1500 citations. Urquhart (2017) is the least-cited article among the top ten, with 376 citations as of April 20, 2023.

No.	Journal	No. of articles	Percentage
1	Finance Research Letters	48	29.27
2	Research in International Business and Finance	19	11.58
3	International Review of Financial Analysis	14	8.54
4	Economics Letters	13	7.93
5	Physica A	7	4.27
6	North American Journal of Economics and Finance	6	3.66
7	Journal of International Financial Markets, Institutions & Money	6	3.66
8	International Journal of Finance and Economics	4	2.44
9	The Quarterly Review of Economics and Finance	4	2.44
10	Studies in Economics and Finance	4	2.44
11	Journal of Empirical Finance	3	1.83
12	Energy Economics	3	1.83
13	International Review of Economics and Finance	3	1.83
14	Other journals	30	18.28
Total		164	100

Table 1: The distribution of articles across the major thirteen journals in the collection (Source: Authors)

Table 2: Top 10 cited papers in the collection as of April 20, 2023 (Source: Authors)

Figure 2 exhibits the percentage of journals and articles in the field belonging to each ABS journal rating between 2016 and December 2022. The left side of the figure indicates that 3* journals are the largest research producers in the field, with 60% of the total number of journals, followed by 2* journals with 33%. The 1* journals are the smallest research producers in the field, with only 7% of the total number of journals. Meanwhile, other ratings, 4 and 4*, are absent because we couldn't find any relevant papers in these categories. Similarly, the right side indicates the percentage of articles rather than journals. This part exhibits that 2* journal articles account for 66% of the total articles in the field, while 3* journal articles account for 30% of the total articles. The 1* journal articles account for only 4% of the total articles.

No	Author	Year	No. of citations
1	Nakamoto, Satoshi	2008	25605
2	Dyhrberg, Anne Haubo	2016	1500
3	Bouri, Elie, Peter Molnár, Georges Azzi, David Roubaud, and Lars Ivar Hagfors	2017	1100
4	Corbet, Shaen, Andrew Meegan, Charles Larkin, Brian Lucey, and Larisa Yarovaya	2018	1027
5	Klein, Tony , Hien Pham Thu, and Thomas Walther	2018	677
6	Bouri, Elie, Rangan Gupta, Aviral Kumar Tiwari, and David Roubaud.	2017	641
7	Bariviera, Aurelio F	2017	590
8	Fry, John, and Eng-Tuck Cheah	2016	521
9	Demir, Ender, Giray Gozgor, Chi Keung Marco Lau, and Samuel A. Vigne	2018	494
10	Ji, Qiang, Bouri, Elie, Lau, Chi, and Roubaud, David	2019	376

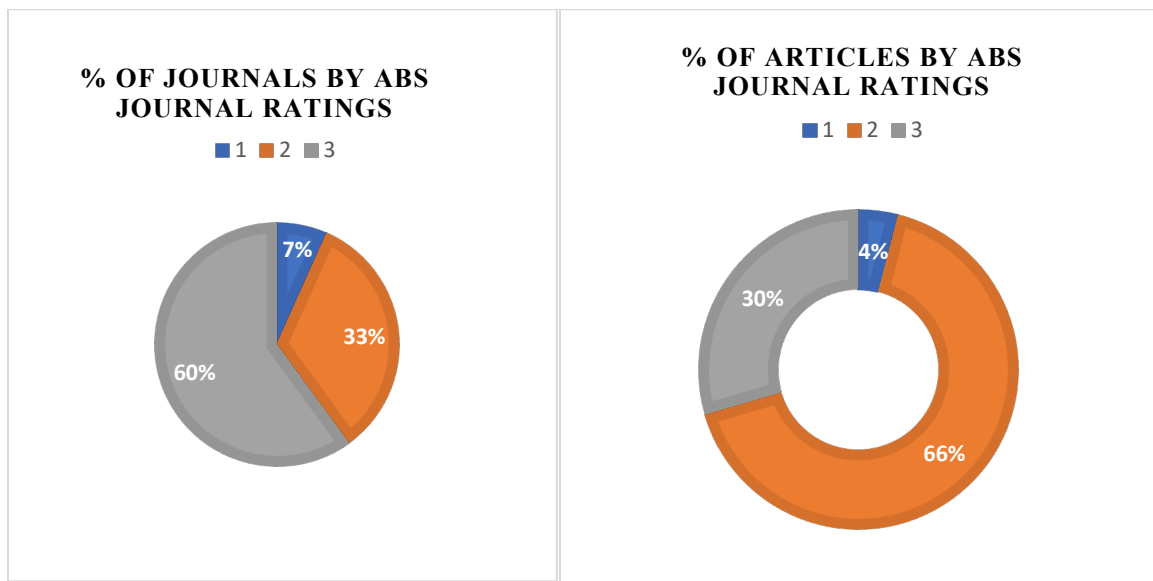


Figure 2. The percentage of journals and articles in the field belonging to each ABS journal rating between 2016 and December 2022

(Source: Authors)

Figure 3 exhibits the top 15 publishing journals in the field between 2016 and December 2022. The total number of articles amounts to 376 journal articles between 2016 and 2022. Finance Research Letters topped the list with 147 articles. It alone accounts for up to 39% of the total scientific production of the top 15 journals. Research in International Business and Finance and Economics Letters are the second-largest scientific producers in the field, with 12.5% and 8.5%, respectively (47 and 34 articles, respectively), of the total production of the top-listed journals. The European Financial Management Journal and the Journal of Financial Stability are the smallest scientific producers on the list, with 0.2% and 0.5%, respectively (1 and 2 articles, respectively).

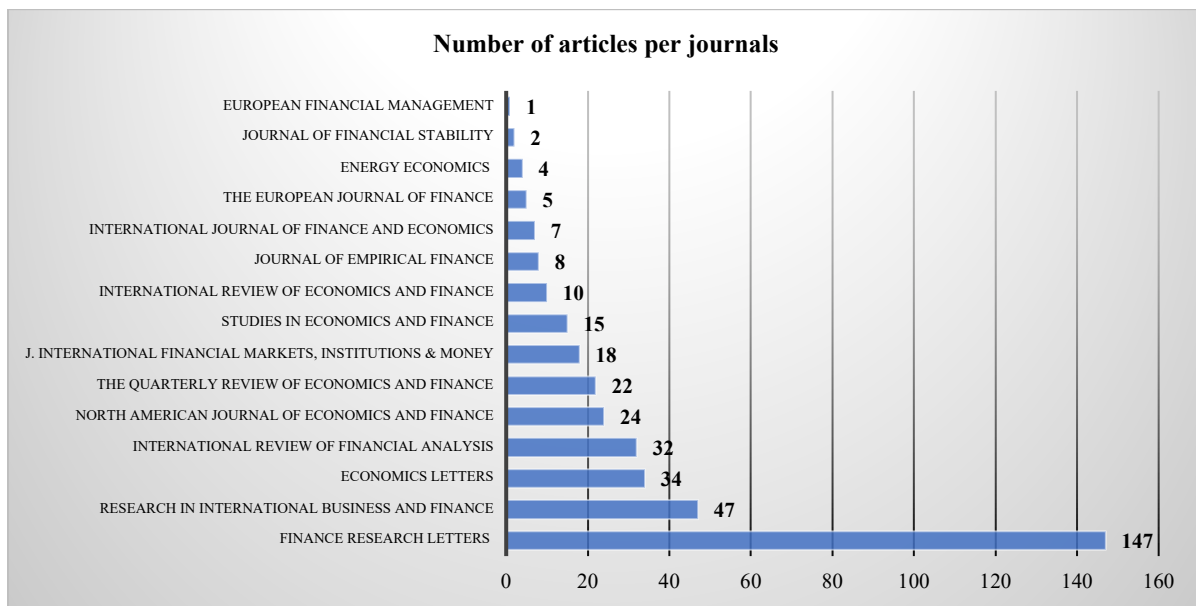


Figure 3. Top publishing journals in the field between 2016 and December 2022

(Source: Authors)

Figure 4 exhibits the evolution of scientific production over time. It starts with three articles in 2016 and grows to seven articles in 2017 and to 25 articles in 2018. Then, scientific production is flat at 56 articles each in 2019 and 2020. After that, it shows a significant increase in 2021, reaching 89 articles. Finally, 144 articles were produced in 2022.

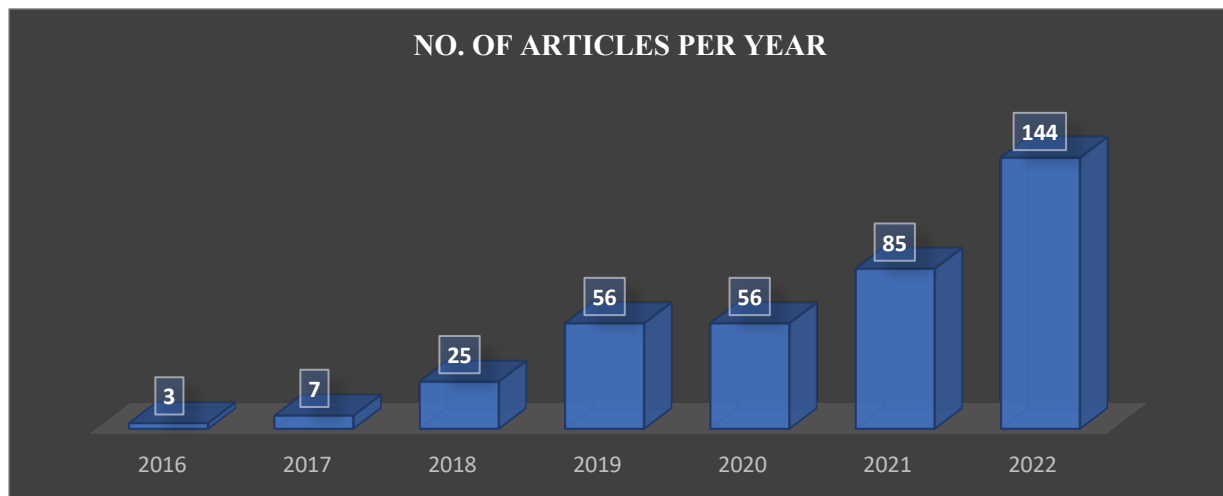


Figure 4. The evolution of scientific production of the top 12 articles from 2016 to December 2022.

(Source: Authors)

3- Research Streams of the Cryptocurrency Volatility Literature

Here, we divide the review into five parts. The first part discusses the evidence on realized volatility. The second part discussed evidence on a wide range of stylized facts, namely, volatility clustering, volatility persistence, asymmetric volatility, leverage effects, volatility spillover, mean reversion, structural breaks, extreme volatility, and tail dependence. The third part discusses the evidence on implied volatility. The fourth part discusses the evidence on stochastic volatility. The fifth part discusses the evidence on the drivers of volatility. This categorization assists in organizing the available knowledge on cryptocurrency volatility, which improving the understanding of the reader by forming a mindmap of the cryptocurrency volatility theory in the literature. Moreover, thos categorization provide the future researchers with a well-organized knowledge of these substantially different lines of research. It is worth mentioning that this paper is benefited significantly from the pioneering work of Poon & Granger (2003) in dividing the relevant sub-topics of volatility and in classifying the relevant stylized facts.

3.1- Realized Volatility

This is also known as historical volatility or realized variance because it represents the actual fluctuations in asset prices over a specific historical period. While cryptocurrencies were introduced in 2009, the examination of realized volatility has substantially grown within the past few years. The oldest papers in the collection of papers included in this review were published in 2017 and written by Estrada (2017) and Pichl & Kaizoji (2017), and the vast majority of papers were published during the past two years. Estrada (2017) examined the causality relationships between Bitcoin and S&P 500 and between Bitcoin and the VIX index. He documented a bidirectional causal relationship between Bitcoin and VIX. The relationship between Bitcoin and the S&P 500, however, was unidirectional from Bitcoin to S&P 500, or the realized volatility of Bitcoin causes the volatility in S&P 500. Pichl and Kaizoji (2017) found that Bitcoin prices are far more volatile compared to the prominent currencies of USD and EUR.

The use of high-frequency data characterizes the literature on cryptocurrency volatility, ranging from daily data (Estrada, 2017; Miglietti, Kubosova, & Skulanova, 2020; Baur & Dimpf, 2021), to hourly data (Ji, Bouri, Kristoufek, & Lucey, 2021), to minute data (Hansena, Kim, & Kimbrough, 2021; Kim, Lee, & Kang, 2020), to second data (Alexander, Heck, & Kaeck, 2021), except for Aalborg, Molnár, and Vries (2019), who used weekly and monthly data in addition to daily data. The use of high-frequency data helps not only overcome the drawback of insensitivity

to the short-term intraday behavior that characterizes traditional volatility models (Corsi, Zumbach, Ulrich, & Dacorogna, 2001), but also improves the accuracy of forecasting and enhances our understanding of the dynamic properties of volatility (Hansen & Lunde, 2011).

It is also noted that centralized markets, especially Coinbase and Binance, have received significant attention from the finance literature, while other centralized and decentralized markets, such as Uniswap, have been neglected. This may be because Coinbase (formerly GDAX) is based in the USA and is widely considered the most popular cryptomarket, while Binance is based in Hong Kong and has the highest trading volume ever. Moreover, all papers document the unexpectedly high volatility of cryptomarkets. For instance, cryptomarkets are more volatile than widely common currencies such as USD and EUR (Miglietti, Kubosova, & Skulanova, 2020; Baur & Dimpf, 2021). As a result, they are recommended to be used as currencies (Baur and Dimpf, 2021). Finally, the introduction of futures increases volatility in the short term and decreases it in the long term (Kim, Lee, & Kang, 2020).

It is noted that only two papers address idiosyncratic volatility in cryptocurrency markets. Zhang and Li (2020) found a positive and robust relationship between idiosyncratic risk and expected returns. This finding contradicts the well-known negative relationship between idiosyncratic risk and expected returns in equity markets. This means that holding high-idiosyncratic volatility cryptocurrencies can enable cryptocurrency investors to earn more profits. In other words, the market will compensate investors for bearing more idiosyncratic risk. In the same vein, Bouri, Kristoufek, Ahmad, & Shahzad (2022) find that the idiosyncratic volatility of cryptocurrencies is priced. In the case of cryptocurrency markets, much room is available for future research in idiosyncratic volatility. For instance, little is known about the determinants of idiosyncratic volatility on cryptocurrency markets. In the same context, the impact of public news arrival and its sentiment on the relationship between idiosyncratic volatility and expected return could be an interesting avenue for future researchers. In addition, the relationship between idiosyncratic volatility and Bitcoin momentum trading is obscure. To the best of our knowledge, no attempts have been made to discover the relationship between herding behavior and idiosyncratic volatility in cryptocurrency markets. Finally, the relationship between idiosyncratic volatility and market liquidity should receive some attention from future researchers.

3.2- Stylized facts

Many definitions of stylized facts have been introduced in the literature. According to Poon and Granger (2003), stylized facts can be defined as common phenomena in financial markets that characterize financial time series and shape market volatility, such as volatility clustering, volatility persistence, asymmetric volatility, mean reversion, volatility spillover, and fat tails. Helfat (2007) defined stylized facts as “observations that have been made in so many contexts that they are widely understood to be empirical truths, to which theories must fit”. Therefore, they are widely understood through the accumulation of evidence that signifies empirical regularities and, in turn, can be considered empirical truths. This leads us to a broader, better, and more generalized understanding of the world. Hirschman (2016) defined stylized facts as some empirical regularities that require sensible explanations. Each will be addressed in detail in the following:

The first stylized fact considered in this paper is *volatility clustering*. We employed the definition of Mandelbrot B. (1963), “large changes tend to be followed by large changes of either sign, and small changes tend to be followed by small changes”. Several papers in the literature provide evidence of volatility clustering in cryptocurrency markets. The first two papers proving volatility clustering in cryptocurrency markets were published in 2016 by Bouoiyour and Selmi (2016) and Dyhrberg (2016). Moreover, it is noticed in the literature that data were collected from several sources, including blockchain.info in Bouoiyour and Selmi (2016), coindesk.com in Dyhrberg (2016), CryptoCompare.com in Gradojevic and Tsiakas (2021), bitcoincharts.com in Naimy and Hayek (2018) and Urquhart (2017), and coinmarketcap.com in Tan, Chan, and Ng (2020), Hafner (2020), and Katsiampa (2019b). It is also noticed that the vast majority of data used in the literature are daily data, except for two studies: Katsiampa, Corbet, and Lucey (2019) relied on hourly data, and Gradojevic and Tsiakas (2021) relied on a one-minute data frequency.

From the discussion above, it is clear that there is a lack of attention to volatility clustering using intraday data (i.e, hourly and minutely data), while there is a complete absence of studies on volatility clustering using the highest frequency of data (i.e., second-by-second data). Therefore, we suggest that future researchers take on this task to enable us to explore common beliefs and well-established theories about the volatility clustering behavior of cryptocurrency markets. This is expected to provide practitioners and policymakers with new information and insights into the intraday dynamics and behaviors of cryptocurrencies.

The previous studies improving our understanding have offered some knowledge on volatility clustering in cryptocurrency markets. So far, we know that volatility clustering exists in

all cryptocurrency markets (Cheikh, Zaied, & Chevallier, 2020; Klein, Thu, & Walther, 2018; Dyhrberg, 2016; Segnon & Bekiros, 2019; Katsiampa, 2019b) based on various GARCH-family models (Bouoiyour & Selmi, 2016) and among the major eight cryptocurrencies using hourly data (i.e., Bitcoin, Ethereum, Litecoin, Dash, Ethereum Classic, Monero, Neo, and OmiseGo) (Katsiampa, Corbet, & Lucey, 2019). Moreover, Bitcoin Cash experiences the strongest volatility clustering behavior, followed by Ethereum, while Bitcoin experiences the weakest (Tan, Chan, & Ng, 2020). In the same context, Gradojevic and Tsiakas (2021) demonstrated that volatility clustering exists in both the time domain and frequency domain using wavelet analysis. Finally, Fakhfekh and Jeribi (2020) and Urquhart (2017) found that volatility clustering exists around whole numbers, with over 10% of prices ending with decimal digits of 00.

Although several studies have focused on volatility clustering, there are many research gaps in the literature. For instance, none in the literature pays attention to the structure of volatility clustering or whether volatility clustering follows nonlinear and asymmetric behavior or not using the copula approach of Chen and Fan (2006). This nonlinear asymmetric volatility clustering can be defined as the asymmetric behavior in the frequency of clusters of low volatilities and high volatilities (Ning, Xu, & Wirjanto, 2015). Second, while volatility clustering is well documented in the literature on cryptocurrency volatility, little is known about the economic rationale that may explain the sources of this phenomenon and the role played by investor behavior and processes of news arrival in inducing volatility clustering. Agent-based models are the most commonly used technique to describe the behavior of asset prices with respect to simple, behavioral, and stylized regulations of investors and other market participants. Agent-based modeling tends to describe financial markets as systematic collections and interactions between autonomous decision-makers, i.e., agents, to induce the behavior of market prices very close to the prices observed in the real world (Cont, 2007). Moreover, research on to what extent herding behavior can induce volatility clustering is required. Schmitt and Westerhoff (2017) introduced a model in which noise traders imitate a linear mix of technical and fundamental strategies to make their decisions. This imitation usually leads to the emergence of volatility clustering. Third, another venue for future research is the usage of entropy measures (i.e., Shannon entropy, Renyi entropy, and Tsallis entropy) for modeling volatility because GARCH-family models have failed to provide a good description of the intraday volatility in financial time series (Bordignon, Caporin, & Lisi, 2007). Fourth, the use of mixed GARCH-jump modeling is also useful for describing discrete financial time series. These models are beneficial in incorporating jumps in asset returns and volatility, and they are better than

the GARCH-multiplier alone in describing the dynamics, especially the financial crashes (Daal, Naka, & Yu, 2007).

The second stylized fact considered in this paper is *volatility persistence*. It first refers to the impact of past or present volatility on future volatility. In other words, volatility persists if future volatility is affected by past or present volatility (Ahmed & Alhadab, 2020). According to Klein, Thu, and Walther (2018), a long memory of financial time series is similar to persistence and can be defined as a long-lived and gradually declining autocorrelation in conditional asset returns or asset volatility. The importance of long memory for asset returns was first proposed by Mandelbrot (1971), who found long-term dependence among a significant portion of a sample of 200 stocks taken from the NYSE. A very slow decay in the autocorrelation of absolute and squared asset return time series is what we mean by the long-memory feature of asset returns. Many empirical papers document this feature in asset returns (Bollerslev & Jubinski, 1999; Bollerslev & Mikkelsen, 1996). Prior research on volatility persistence can be divided into several pathways. The first pathway focuses on the existence of volatility persistence. Some research has discovered high persistence in cryptocurrency markets (Cheikh, Zaied, & Chevallier, 2020; Segnon & Bekiros, 2019; Katsiampa, Corbet, & Lucey, 2019; Katsiampa, 2019b; Liu & Serletis, 2019). Other research documented the existence of volatility persistence, but the results did not indicate high levels of it (Tan, Chan, & Ng, 2020; Omane-Adjepong, Alagidede, & Akosah, 2019; Abakah, Gil-Alana, Madigu, & Romero-Rojo, 2020; Mensi, Al-Yahyaee, & Kang, 2019).

The second pathway focuses on the similarity between cryptocurrencies and commodities and other financial assets in terms of volatility persistence (Klein, Thu, & Walther, 2018; Dyhrberg, 2016). Klein, Thu, and Walther (2018) used daily data from July 1, 2011 to December 31, 2017 to compare the strength of persistence between Bitcoin and commodities. Their findings indicated significant long memory in the Bitcoin time series, but the strength of persistence is not as strong as for gold and silver. They attributed the relative weakness of persistence in Bitcoin to the recentness and short history of cryptocurrencies in general and Bitcoin in particular. In contrast, Dyhrberg (2016) demonstrated that Bitcoin is the same as gold in terms of high volatility persistence and its sensitivity to changes in exchange rates, while Bitcoin is the same as the US dollar in terms of its capability to be used as a medium of exchange as they respond similarly to federal fund rates.

The third pathway focuses on the persistence of volatility in different market conditions (Bouoiyour & Selmi, 2016; Fakhfekh & Jeribi, 2020; Yaya, Ogbonna, & Mudida, 2021; Mnif,

Jarboui, & Mouakhar, 2020). Bouoiyour and Selmi (2016) studied two time periods: from December 2010 to December 2014, when Bitcoin showed large volatility, and from January 2015 to July 2016, when the volatility of Bitcoin decreased significantly. The former showed strong persistence and explosive behavior, while the latter showed significant persistence but was far from being considered a long-memory process. Yaya, Ogbonna, and Mudida (2021) studied the volatility persistence of 12 cryptocurrencies, namely Bitcoin, Dash, Digibyte, Doge, Ethereum, Litecoin, Maidsafecoin, Monero, Nem, Ripple, Stellar, and Vertcoin, during the pre-and post-crash periods of Bitcoin, i.e. August 7, 2015 and November 28, 2018. They revealed that volatility persistence tends to exist for shorter periods in post-crash periods than in pre-crash periods. Fakhfekh and Jeribi (2020) showed that volatility persistence is stronger in bull markets compared to bear markets. Finally, Mnif, Jarboui, and Mouakhar (2020) studied the volatility persistence proxied by the Hurst exponent before and after the COVID-19 pandemic. The results showed that the Hurst exponent amounts to 0.40, indicating that the five top cryptocurrency markets, i.e., Bitcoin, Ethereum, Ripple, Litecoin, and Binance, follow anti-persistent behavior in both cases.

The fourth and last pathway focuses on the use of re-scale range analysis, detrended fluctuation analysis, fractional integrated GARCH (FIGARCH) and its extensions, and wavelet analysis to detect persistence and long memory in cryptocurrency volatility. Almost all papers in this review evidenced the existence of persistence and long memory in cryptocurrency volatility (Chaim & Laurini, 2019; Phillip, Chan, & Peiris, 2018; Celeste, Corbet, & Gurdgiev, 2020; Rambaccussing & Mazibas, 2020; Al-Yahyaee, Mensi, Ko, & Yoon, 2020; Mensi, Lee, Al-Yahyaee, Sensoy, & Yoon, 2019; Bariviera, 2017). The only exception was Takaishi (2020), who documented anti-persistent and rough volatility in the Bitcoin market using daily data from January 28, 2015 to January 6, 2019.

Due to the relative recentness of cryptocurrencies, many gaps exist in the current body of knowledge. For instance, very little is known about the long-memory feature in cryptocurrency markets using intraday data. In other words, little is known about the intraday volatility behavior, dynamics, or stylized facts in cryptocurrency markets. Moreover, future research should pay attention to the differences in the evolution and presence of persistence and long memory among the major cryptocurrency markets. In addition, a much-needed direction for future research is to look at the drivers and economic rationale of differences in the volatility dynamics of different cryptocurrency markets. Furthermore, we need to examine the impact of sudden changes or sudden shocks on volatility persistence in cryptocurrency markets by using the ICSS algorithm proposed by Inclan and Tiao (1994). The

relationship between public information, proxied by the number of daily economic news headlines, and volatility persistence is also an interesting area for future research¹. Finally, examining persistence and long memory in cryptocurrency markets after adjusting GARCH-family models to account for time-varying unconditional means is also a good avenue for future research.²

The third stylized fact considered in this paper is *asymmetric volatility*. It is well known that asymmetric volatility means a different response to bad news and good news. In particular, negative shocks have a larger impact on volatility than positive shocks of the same magnitude. One interpretation of this phenomenon is that investors react more quickly to bad news than to good news, which contributes more volatility to the market (Jones, Walker, & Wilson, 2004). Prior literature evidenced the existence of asymmetric volatility using daily data (Baur & Dimpfl, 2018; Katsiampa, Corbet, & Lucey, 2019; Cheikh, Zaied, & Chevallier, 2020; Bouoiyour & Selmi, 2016; Klein, Thu, & Walther, 2018; Dyhrberg, 2016; Segnon & Bekiros, 2019; Katsiampa, Corbet, & Lucey, 2019; Naimy & Hayek, 2018; Caporale & Zekokh, 2019; Apergis, 2022; Kakinaka & Umeno, 2022; Mensi, Al-Yahyaee, & Kang, 2019). Only one paper demonstrated asymmetric volatility using intraday data (Mensi, Lee, Al-Yahyaee, Sensoy, & Yoon, 2019). This is an additional indication of the lack of attention paid to the dynamics and behavior in the cryptocurrency market.

Conflicting findings are found in the literature with respect to the similarities and differences between cryptocurrencies and other financial and commodity markets. Some scholars documented unique asymmetric volatility, namely an inverted asymmetric response to good and bad news, in cryptocurrency markets, as the variance response to good news is stronger than that to bad news. Similar to gold and silver and different from the stock market, the strong response to good news can be explained by the herding behavior of uninformed investors because some investors buy in response to the fear of missing out on rising cryptocurrency valuations, while the weak response to bad news can be attributed to the contrarian behavior of informed investors (Baur & Dimpfl, 2018; Cheikh, Zaied, & Chevallier, 2020; Klein, Thu, & Walther, 2018; Fakhfekh & Jeribi, 2020). Other scholars have discovered that the response of cryptocurrency volatility to market shocks is symmetric rather than asymmetric (Dyhrberg, 2016; Katsiampa, 2019b; Charles & Darné, 2019). The final group documented normal asymmetric volatility behavior with a stronger response of variance to good news compared to bad news (Mensi, Lee, Al-Yahyaee,

¹ For more information, see Janssen (2004).

² For more information, see McMillan & Ruiz (2009).

Sensoy, & Yoon, 2019; Bouoiyour & Selmi, 2016; Mensi, Al-Yahyaee, & Kang, 2019; Segnon & Bekiros, 2019).

With regard to the models, the empirical evidence documents the superiority of some asymmetric models over others. For instance, Klein, Thu, and Walther (2018) recommended fractionally integrated APARCH (FIAPARCH) as the best-fitting model over asymmetric power ARCH (APARCH). Katsiampa, Corbet, and Lucey (2019) found evidence that asymmetric diagonal BEKK is better than diagonal BEKK. Naimy and Hayek (2018) recommended the EGARCH model over the basic GARCH and exponentially weighted moving average (EWMA). Mensi, Al-Yahyaee, and Kang (2019) found evidence that the FIGARCH model with a structural break is better than other models, such as the ARFIMA-GARCH, ARFIMA-FIAPARCH, and ARFIMA-HYGARCH models.

Future research should focus on the intraday data to handle the issue of the large number of observations needed by some important tests. Also, multivariate GARCH models are important to investigate the relationship between Altcoins and Bitcoin. Furthermore, regarding the determinants of asymmetric volatility, Wu (2001) found that the leverage effect and the volatility feedback effect are important determinants of asymmetric volatility in the equity market, while very little is known about the determinants of asymmetric volatility in the cryptocurrency market. In addition, the relationship between news sentiment and asymmetric volatility in the cryptocurrency market should attract the attention of future researchers. Smales (2015) explained that covering this issue should provide an answer to two questions: First, how is the volatility of cryptocurrency returns impacted by the sentiment of non-scheduled news events? Second, how does the state of the business cycle affect the relationship between news and volatility in the cryptocurrency market?

Another interesting area of research is investigating the relationship between trading volume, as a proxy for the arrival of information at the cryptocurrency market, and asymmetric volatility during the COVID-19 pandemic using an asymmetric GARCH model such as EGARCH. Finally, we need to examine asymmetric herding, asymmetric attention, differences of opinions, and loss aversion as possible behavioral explanations for asymmetric volatility in the cryptocurrency market.

The fourth stylized fact considered in this paper is *the leverage effect*. It is considered to be the key source of asymmetric volatility. It can be defined as the well-known negative relationship between asset returns and volatility (Black, 1976; Christie, 1982). In other words, in

the case of equities, a decline in stock value leads to an increase in the firm's leverage, as measured by the debt-to-equity ratio. Stockholders or investors will accordingly evaluate the future cash flow of the firm as being riskier. The prior literature can be divided into two groups, as follows:

The first group of studies focused on whether the leverage effect exists or not. In this context, some studies have evidenced the existence of the leverage effect in cryptocurrency markets. For instance, Phillip, Chan, and Peiris (2018), Caporale and Zekokh (2019), and Bouoiyour and Selmi (2016) found a negative relationship between one-day ahead asset returns and volatility. Because the cryptocurrency market works 24 hours a day, 7 days a week, it is suggested that the prices at midnight can be considered closing prices (Catania & Grassi, 2022). Corbet and Katsiampa (2020) demonstrated that the leverage effect exists using minutely, hourly, daily, and weekly data from June 2010 to February 2018. Finally, Liu and Serletis (2019) found a leverage effect in Bitcoin but failed to find it in Ethereum. Some other scholars discovered an inverted leverage effect. In other words, they found a positive rather than a negative relationship between asset returns and volatility (Cheikh, Zaied, & Chevallier, 2020; Klein, Thu, & Walther, 2018; Mensi, Al-Yahyaee, & Kang, 2019; Fakhfekh & Jeribi, 2020). Finally, Naimy and Hayek (2018) and Dyhrberg (2016) found an insignificant leverage effect, i.e., no relationship between asset returns and volatility. This insignificant relationship means that neither good nor bad news influences asset volatility asymmetrically. Panagiotidis, Papapanagiotou, & Stengos (2022) and Fung, Jeong, & Pereira (2022) find that an invested leverage effect exists in the majority of the 292 and 254 cryptocurrencies that constitute their sample respectively. Huang, Ni, & Xu (2022) show that the presence of leverage effect or inverted leverage effect in cryptocurrencies relies to a great extent on both the jump and diffusive components of the correlation between return and volatility.

The second strand of literature paid attention to the contribution of the leverage effect to the predictability of cryptocurrency volatility. Yu (2019) used a 5-minute data frequency to examine the impact of the leverage effect and EPU on the predictability of Bitcoin volatility. Employing HAR-family models, his findings showed that not only does the leverage effect carry beneficial predictive information in predicting Bitcoin volatility, but also that its predictability power is stronger than the jump component. As a result, both the leverage effect and EPU enhance the predictability power of HAR-family models. Klein, Thu, and Walther (2018) used a sample of 102 active cryptocurrencies to investigate the impact of the leverage effect and volatility persistence on the predictability of cryptocurrency volatility. The use of Garman and Klass's

volatility measures indicates that both the leverage effect and volatility persistence can play a useful role in predicting cryptocurrency volatility as well as reducing risk and speculation in cryptocurrency markets.

Future recommendations may include the role of the leverage effect and investor sentiment in forecasting cryptocurrency volatility using HAR-family models. To do so, investor sentiment and the leverage effect need to be incorporated into the basic HAR model to answer the following questions:

- Does investor sentiment or the leverage effect contain more information?
- Do investor sentiment and leverage effect improve the predictability of cryptocurrency volatility?

Another interesting way is to combine the leverage effect with EPU to investigate their impact on the predictability of cryptocurrency volatility within the context of regime switching. In the case of equity markets, the HAR-RV models containing the leverage effect and EPU with regimes make significant improvements in forecasting accuracy (Duan, Chen, Zeng, & Liu, 2018). Also, the use of static and dynamic copula models has not been given sufficient attention in the current state of the-art. This kind of model can not only help isolate the leverage effect from the marginal distribution of asset returns and volatility but also provide us with relevant information about the structure of the leverage effect (Ning, Xu, & Wirjanto, 2008). In the same context, the copula technique can be merged with the DCC model to estimate the leverage effect and time-varying correlations between cryptocurrency markets and other financial and commodity markets. This approach is beneficial for checking whether there is a hedging possibility between any pair of markets. Finally, the COVID-19 pandemic opened a new venue for future research. Future research can study the impact of the pandemic on the leverage effect using daily and intraday data by employing efficient GARCH-family models, such as fractionally integrated exponential GARCH-in-mean (FIEGARCH-M).

The fifth stylized fact considered in this paper is *volatility spillover*. Volatility spillover is the transmission of volatility between different assets and different financial markets. In other words, it can be defined as how the volatility of an asset or a market impacts or is impacted by the volatility of another asset or another market (Poon and Granger, 2003). Previous related literature has shown interest in volatility spillover between major cryptocurrencies (Katsiampa, 2019a,b; Ji, Bouri, Lau, & Roubaud, 2019; Katsiampa, Corbet, & Lucey, 2019), cryptocurrency markets and commodities such as gold (Mariana, Ekaputra, & Husodo, 2021) and crude oil (Okorie & Lin,

2020), cryptocurrencies and financial markets (Mariana, Ekaputra, & Husodo, 2021), and cryptocurrencies and the stocks of technology firms (Symitsi & Chalvatzis, 2018). A discussion of each is provided in the following:

First, several papers address the volatility spillover among the major cryptocurrencies. For instance, Katsiampa (2019a) evidenced the existence of comovement among the five major cryptocurrencies, namely, Bitcoin, Ether, Ripple, Litecoin, and Stellar Lumen. He found empirical evidence that the conditional covariances are impacted by the cross-product of previous error terms and previous covariance terms. Katsiampa, Corbet, and Lucey (2019) supported the previous findings using hourly data. They demonstrated that there is strongly positive interdependence between the eight major cryptocurrencies, i.e., Bitcoin, Ethereum, Litecoin, Dash, Ethereum Classic, Monero, Neo, and OmiseGO. Beneki, Koulis, Kyriazis, and Papadamou (2019) showed that there is a one-way transmission of information and volatility from Ethereum to Bitcoin using BEKK-GARCH and the impulse response function. Fasanya, Oyewole, and Odudu (2021), Ji, Bouri, Lau, and Roubaud (2019), Yi, Xu, and Wang (2018), and Qiao, Zhu, and Hau (2020) supported the interdependence and volatility among major cryptocurrencies and confirmed that Bitcoin has the largest impact on the volatility of other cryptocurrency markets. In addition to the previous studies, Kumar and Anandarao (2019) documented that the volatility spillover was small before 2017 and increased significantly after that, indicating that this amount of volatility transmission across cryptocurrency markets enables investors to benefit from the available possibilities of diversification. In the same context, Omane-Adjepong, Alagidede, and Akosah (2019) demonstrated that the volatility spillovers among the major cryptocurrency markets are sensitive to trading scales. Finally, Koutmos (2018) used volatility spillover as a proxy for contagion risk and confirmed that cryptocurrency markets are becoming more integrated over time, meaning that volatility spillover has increased with the passing of time. Kumar, Iqbal, Mitra, Kristoufek, & Bouri (2022) demonstrate that the volatility spillover among the major cryptocurrencies is stronger during the COVID-19 pandemic than before it. In the same vein, Ozdemir (2022) shows that Bitcoin, Ethereum, and Litecoin were mutually dependent during the COVID-19 pandemic. He also demonstrates that the volatility spillover was stronger during the second lockdown of the COVID-19 pandemic. Future research recommendations may take a look at exploring and quantifying the principal transmitter or receiver for a totality of cryptocurrency markets over various time spans and also over various horizons of investment. Moreover, new research is required to investigate the interdependence between price volatility and liquidity

volatility in order to improve our understanding of the dynamics of cryptocurrency markets, especially in light of the growing evidence of manipulations and other trading idiosyncrasies. Finally, we found initial evidence that cryptocurrency markets are becoming more integrated over time. This evidence was based on the major 18 cryptocurrency markets. More research is needed to provide some support for this initial finding with a wider range of cryptocurrency markets and various data frequencies.

The second stream of research focused on the volatility spillover between cryptocurrency markets and other financial markets. Symitsi and Chalvatzis (2018) found bidirectional volatility spillover between financial stock indices and Bitcoin, short-term volatility spillover from technology stocks to Bitcoin, and long-term volatility spillover from Bitcoin to energy stocks. Alexander, Heck, and Kaeck (2021) used second-by-second data and evidenced that there is a strong volatility spillover between the major cryptocurrency markets and the US stock market during the trading hours and a weak spillover between them during the closing hours. Given the decreasing gravity of bitcoin in the cryptocurrency markets, future researchers may replicate the previous study to investigate the volatility spillover within Ethereum spots and perpetuals on various cryptocurrency exchanges, examine Ripple and other prime cryptocurrencies, or investigate the volatility spillover between Bitcoin and Ethereum using spot and perpetual instruments. Liu and Serletis (2019) documented strong volatility spillover between the major cryptocurrency markets and the US, UK, German, and Japanese stock markets. They also provided evidence that the more acceptable and used the cryptocurrencies, the stronger the volatility spillover. Finally, Corbet, Larkin, Lucey, Meegan, and Yarovaya, (2020) found that mineable cryptocurrencies are more sensitive to US monetary policy volatility spillover and feedback compared to those that are non-mineable. An interesting area for future research is to investigate the volatility spillover between cryptocurrency markets and other financial markets, such as bond markets, derivatives markets, and currency markets, between cryptocurrency markets and real estate markets, between cryptocurrency markets and indices of emerging markets, between cryptocurrency markets and indices of Islamic financial markets, and between cryptocurrency markets and private equity markets. Naeem, Lucey, Karim, & Ghafoor (2022) discover a unique spillover between the volatility of financial assets and cryptocurrency indices. Harb, Bassil, Kassamany, & Baz (2022) find that there is no volatility spillover between cryptocurrencies and conventional financial assets (i.e., stocks and bonds). Cao & Xie (2022) demonstrate a strong asymmetric volatility spillover between China's financial market and cryptocurrency.

The third group of scholars examined the volatility spillover between cryptocurrency markets and commodities markets. Fang, Bouri, Gupta, and Roubaud (2019) found a positive impact of the global EPU on the Bitcoin-commodities spillover. Ji, Bouri, Lau, and Roubaud (2019) showed that gold prices have a positive impact on the net directional volatility spillover. The US uncertainty, on the other hand, has a negative impact on the net directional-volatility spillover. Mariana, Ekaputra, and Husodo (2021) documented the existence of volatility spillover between gold and both Bitcoin and Ethereum. Okorie and Lin (2020) evidenced a unidirectional volatility spillover from crude oil to cryptocurrency markets. Attarzadeh & Balcilar (2022) show that clean energy receives volatility shocks from Bitcoin. Pham, Nguyen, & Do (2022) document a volatility spillover from conventional cryptocurrencies to thermal coal futures. Future research may shed light on the volatility spillover between cryptocurrency markets and agriculture commodity markets, between cryptocurrency markets and carbon markets, and between cryptocurrency markets and other energy (fossil) markets, such as natural gas, coal, and Brent oil.

The sixth stylized fact considered in this paper is *mean reversion*. Mean reversion refers to the tendency of asset price volatility or past returns to fluctuate in the short term and to return to the long-term trend or fundamental level's sample average (Poon, 1996). Several papers have evidenced the presence of mean-reversion behavior in cryptocurrency markets (Leung & Nguyen, 2019; Aslan & Sensoy, 2020; Naimy & Hayek, 2018; Lahmiri, Bekiros, & Salvi, 2018; Borgards & Czudaj, 2021). In contrast, few papers contradicted the presence of mean-reversion in these markets. For instance, Gil-Alana, Abakah, and Rojo (2020) used a sample period from May 7, 2015, to October 5, 2018, with 826 trading days, to investigate the existence of mean-reversion in the six largest cryptocurrency markets, namely, Bitcoin, Ethereum, Ripple, Litecoin, Stellar, and Tether. The findings show that only Tether follows mean-reversion behavior. Caporale, Gil-Alana, and Plastun (2018) failed to find mean-reversion in the four major cryptocurrency markets, namely, Bitcoin, Litecoin, Ripple, and Dash, from 2013 to 2017. Similarly, Turatti, Mendes, and Caldeira (2020) could not discover mean-reversion behavior using weekly data from 2010 to 2019.

Other key findings in the literature include asymmetric mean-reversion behavior that entails a more powerful and speedier reversion of negative prices relative to positive ones using a wide range of data frequencies, namely, minutely, hourly, daily, and weekly (Corbet & Katsiampa, 2020). Furthermore, mean-reversion behavior climbed significantly after the price boom of 2017 compared to before that date (Bouri, Vo, and Saeed, 2021). Yarovaya, Matkovskyy, and Jalan

(2022) discovered mean-reversion behavior only at the lower quantiles (from the lowest to 0.30), while other quantiles tend to show high persistence, leading to long-memory behavior.

Theoretically, we attribute the mean-reversion behavior to the overreaction hypothesis. This entails that buying past losers and short-selling past winners may lead to abnormal returns in the short term. To be more specific, when new information arrives, some investors tend to initially overreact and deviate the prices from the fundamental values. After a short period, the prices will be corrected by returning to their intrinsic values (De Bondt & Thaler, 1985). Future research may shed some light on the underlying source of the underreaction by investigating the order limit book's supply and demand dynamics during the overreaction. Also, analyzing these phenomena in other asset classes, such as the Bitcoin spot market, Bitcoin futures, and options, would be an interesting area for future research.

With the dominant impact of COVID-19 on all aspects of our lives, Yarovaya, Matkovskyy, and Jalan (2022) provided evidence that cryptocurrencies are not only the riskiest and highest financial asset in terms of recovery rates and mean-reversion but are also highly persistent assets. In this context, future researchers may take up the task of examining the quickness of half-lives mean-reversion.

The seventh stylized fact considered in this paper is *structural breaks*. Structural breaks can be defined as long-term changes and/or momentary jumps stemming from policy, technological, or institutional transformations, natural disasters, significant movements in investor sentiment, or large economic fluctuations such as large changes in oil prices, leading to a jump or break in the parameters (Li, Cheng, & Fang, 2020; Narayan & Smyth, 2015). The relevant literature on structural breaks can be divided into four categories, as follows:

First, some studies focused on the inclusion of structural breaks in the prediction model and their impact on forecasting accuracy. The evidence provided by these papers indicates that accounting for structural breaks improves the forecasting process. As a result, prediction models that account for structural breaks outperform those that do not (Mensi, Al-Yahyaee, & Kang, 2019; Shen, Urquhart, & Wang, 2020; Mba & Mwambi, 2020; Charles & Darné, 2019; Hu, Valera, & Oxley, 2019).

The second category paid attention to the existence of structural breaks in the cryptocurrency markets. Conflicting findings have been found in the prior literature. For instance, Katsiampa (2019b) found strong evidence of the existence of structural breaks in conditional variance in Bitcoin and Litecoin but could not find them in Either, Ripple and Stellar Lumen.

Charfeddine and Maouchi (2019) found evidence of the existence of structural breaks in three out of four cryptocurrencies, namely, Bitcoin, Ripple, and Litecoin, while structural breaks could not be found in Ethereum. In line with the previous paper, Gemici and Polat (2021) also failed to find structural breaks in Ethereum, while they did find structural breaks in Bitcoin and Litecoin. Katsiampa, Moutsianas, and Urquhart (2019), Mensi, Lee, Al-Yahyaee, Sensoy, and Yoon (2019), and Wang, Wang, Yin, and Ji (2021), on the other hand, discovered structural breaks in Bitcoin and Ethereum from December 17, 2017 to January 14, 2018, from June 1, 2013 to June 23, 2018, and from March 15, 2017 to April 30, 2020 respectively. Thies and Molnár (2018) and Matkovskyy and Jalan (2017) found strong evidence in favor of the existence of structural breaks in Bitcoin from September 2011 to August 2017 and from April 2015 to October 2018, respectively. Bouri, Gupta, and Roubaud (2019) documented the existence of structural breaks and nonlinearities in the largest 14 cryptocurrencies, namely, Bitcoin, Ethereum, Ripple, Litecoin, Stellar, Dash, Nem, Monero, Bytecoin, Verge, Siacoin, BitShares, Decred, and Dogecoin. This existence implies that the use of time-varying models is essential because the use of static models leads to incorrect conclusions. Palamalai, Kumar, and Maity (2021) documented the existence of structural breaks in the top ten cryptocurrencies in terms of market capitalization.

The third category links structural breaks with periods of economic recession. For instance, James, Menzies, and Chan (2021) proved that all studied cryptocurrencies, i.e., 51 cryptos, experienced similar structural breaks before the COVID-19 pandemic, while they became different during the pandemic. In the same context, Mandaci and Cagli (2022) document structural breaks during the COVID-19 pandemic. More precisely, they found structural breaks in the third quarter of 2020 using intraday data. Gil-Alana, Abakah, and Rojo (2020) investigated the six largest cryptocurrencies in times of financial crises and provided evidence that structural breaks existed during the 2017 financial crisis in all cases. Similarly, Yaya, Ogbonna, Mudida, and Abu (2021) documented the existence of structural breaks in the leading 12 cryptocurrencies during the pre- and post-crash periods of Bitcoin prices on December 17, 2017. Similarly, Bouri, Gil-Alana, Gupta, and Roubaud (2019) documented the existence of structural breaks during the crash period of Bitcoin prices in December 2013.

The fourth category addressed the spillover of structural breaks. Canh, Wongchoti, Thanh, and Thong (2019) found evidence that breaks are transmitted from small market capitalization cryptocurrencies (Dash, Monero, in 2016 and Ripple in April 2017) to larger ones (Litecoin, Stellar, Bytecoin, and Bitcoin in May 2017). Akyildirim, Corbet, Katsiampa, Kellard, and Sensoy

(2020) used minute data and provided evidence on the spillover of structural breaks from the Chicago Board Options Exchange (CBOE) to Chicago Mercantile Exchange (CME) bitcoin futures.

Future research could attempt to jointly study contagion and structural breaks during times of crises and crashes, and there is also a need to investigate contagion and structural breaks jointly in a multivariate setting, as suggested by Matkovskyy & Jalan (2017). Moreover, much research is required with regard to the determinants of liquidity, investor sentiment, and macro-finance and their impact on cryptocurrency prices and volatility (Palamalai, Kumar, & Maity, 2021). In addition, Hu, Valera, and Oxley (2019) recommended future research to examine the bubble spillover between Bitcoin and other major cryptocurrencies. Finally, Mandaci and Cagli (2022) suggested that future researchers investigate the drivers of herding behavior in cryptocurrency markets.

The eighth stylized fact considered in this paper is *extreme volatility*. Extreme volatility can be defined as a period when extremely large volatility is expected. Engel and Hakkio (1993) compared normal volatility and extreme volatility. They defined normal volatility as normal changes or normal increases and decreases in asset prices that usually happen over time, while extreme volatility is unusually greater changes in asset prices that happen rarely. The main stream of research exhibited the extreme volatility feature of cryptocurrency markets (Mba, Pindza, & Koumba, 2018; Gkillas & Katsiampa, 2018; Brauneis & Mestel, 2019; Celeste, Corbet, & Gurdgiev, 2020; Omane-Adjepong & Alagidede, 2019; Phillip, Chan, & Peiris, 2019; Silahli, Dingec, Cifter, & Aydin, 2021; Catania & Grassi, 2022; Qiao, Zhu, & Hau, 2020).

Baur and Hoang (2021) demonstrated that Bitcoin is similar to gold in terms of acting as a safe haven and storing value. However, because Bitcoin shows extreme volatility behavior, investors should use Tether as a protective tool and as a safe haven against the extreme volatility of Bitcoin. Phillip, Chan, and Peiris (2018) studied the five largest cryptocurrencies in terms of market capitalization, namely, Bitcoin, Ethereum, Ripples, Dash, and NEM. Their findings suggest that only Ripple shows extreme volatility. Ji, Bouri, Lau, and Roubaud (2019) suggested that the six largest cryptocurrencies, namely, Bitcoin, Ethereum, Ripple, Litecoin, Stellar, and Dash, can successfully act as investable assets despite their extreme volatility because they generate higher returns. They also suggest that the extremely volatile nature of Bitcoin motivates herding behavior among investors. Mensi, Al-Yahyaee, Al-Jarrah, Vo, and Kang (2020) found that Bitcoin shows

extreme volatility. As a result, investors are advised to increase the proportion of Ethereum, Litecoin, and Ripple in their portfolios at the expense of Bitcoin.

Liu, Semeyutin, Lau, and Gozgor (2020) document short periods of extreme volatility in the cryptocurrency markets using daily data. Hasan, Naeem, Arif, and Yarovaya (2021) used minute data and found evidence that the COVID-19 pandemic increases crash risk and increases the proneness to extreme economic and financial events, resulting in extreme volatility. In the same context, Bouri, Saeed, Vo, and Roubaud (2021) demonstrated that extreme events such as the COVID-19 pandemic lead investors and traders in cryptocurrency markets to pay more attention to extreme events and news that occur in the lower and upper tails. Zhang, Chan, Chu, and Nadarajah (2019) focused on the dominant four cryptocurrencies, namely, Bitcoin, Ethereum, Ripple, and Litecoin, and provided evidence that extreme volatility exists using hourly data.

Very little is known about jump risk in cryptocurrency markets. Therefore, future research may focus on investigating jump risk when modeling volatility, especially as there is convincing empirical evidence that neglecting jump components negatively affects the volatility models and makes them incapable of depicting the real empirical features of the studied time series (Pan, 2002). Moreover, it is recommended that future researchers shed some light on the time-varying feature when studying extreme events. Further, they may consider not only the differences between various cryptocurrencies when studying the extreme volatility but also the key drivers of these differences. In addition, the impact of extreme returns and extreme volatility is still unclear and lacks sufficient attention from academics and practitioners. Finally, extreme asymmetric volatility has not received the attention of scholars.

The ninth stylized fact considered in this paper is *tail dependence*. Tail dependence describes the degree of dependence among extreme events or the degree of dependence in the upper and lower quantiles of the joint distribution (Oordt & Zhou, 2012). The relevant literature with respect to tail dependence in cryptocurrencies can be divided into three separate groups as follows.

The first group focused on the tail dependence among major cryptocurrencies. Borri (2019) used CoVaR and daily data to measure the tail dependence between the four major cryptocurrencies, namely, Bitcoin, Ethereum, Ripple, and Litecoin. His findings indicated that all the abovementioned cryptocurrencies are greatly associated with one another. Similar to the previous study, Tiwari, Adewuyi, Albulescu, and Wohar (2020) provided evidence that heavy upper and lower tail dependence exists between each pair of the studied cryptocurrencies using

non-parametric mixture copulas from August 4, 2013 to June 17, 2018. This high contagion risk indicates that investors should not use diversification to hedge and protect themselves against contagion risk. Also, this large contagion risk may be beneficial for policymakers who are interested in the regulation of cryptomarkets. Shahzad, Bouri, and Ahmad (2022) used the LASSO technique with quantile regression to examine the connectedness among cryptocurrencies and found that small cryptocurrencies, rather than Bitcoin, are the key risk transmitters and receivers. In contrast, Hasan, Naeem, Arif, and Yarovaya (2021) used a network-based approach and found that Bitcoin and Litecoin are the key risk receivers, while Ripple and Binance Coin are the key risk transmitters, using minute data. These findings have some implications for investors and portfolio managers who are interested in developing new trading strategies using the risk and return of cryptomarkets. In addition, investors may need to chase the crash risk and fat-tail risk spillover across the cryptomarkets.

The LASSO technique with quantile regression was also used by Nguyen, Chevapatrakul, and Yao (2020), demonstrated evidenced that the right-tail dependence is more powerful than the left-tail one. As a result, the naïve equally-weighted technique can be used as a diversification tool when investing in cryptomarkets. Furthermore, their findings show that Bitcoin and Litecoin can serve as risk transmitters during times of bull markets, while Ethereum and Ethereum Classic can do this job in times of bear markets. This implies that policymakers and market participants can use these studies to increase their awareness of the sources of destabilization in cryptomarkets, which could destabilize the whole economy. Chaim and Laurini (2019) showed a long-term dependence on crypto return volatility. Bouri, Saeed, Vo, and Roubaud (2021) used quantile-based connectedness measures based on the variance decomposition of a quantile vector autoregression model and provided some interesting findings. First, the connectedness measures demonstrate that shocks spread at the lower and upper tails more strongly than those in the mean and median. Second, the dependence structures at the lower and upper tails are different from those in the mean and median. Zhang, Chan, Chu, and Nadarajah (2019), in their paper, used hourly data instead of daily data. Their findings show that the most powerful lower and upper tail dependence lies between Bitcoin and Ethereum, while the weakest is between Bitcoin and Ripple or Litecoin and Ripple.

Naeem, Bouri, Boako, and Roubaud (2020) used GARCH-copula models to describe the major three cryptocurrencies, namely, Bitcoin, Ethereum, and Litecoin. They found that the tail dependence is more prominent when returns and volume are large, while the dependence is weak

when returns and volume are small. Ahelegbey, Giudici, and Mojtahedi (2021) used the extreme downside hedge (EDH) and extreme downside correlation (EDC) to measure the tail risk. They found that Bitcoin is the key transmitter of tail contagion, while Ethereum is the key receiver. In contrast to the previous paper, Xu, Zhang, and Zhang (2021) demonstrated that Bitcoin is the largest risk receiver while Ethereum is the largest risk transmitter using the TENET approach as a measure of tail dependence.

The second group addressed the tail dependence between cryptocurrencies and other financial markets. Chan, Chu, Zhang, and Nadarajah (2022) investigated the extreme dependence between Bitcoin and Ethereum versus the euro and US dollar using hourly data from July 2017 to April 2020 and employing extreme value theory to analyze the data. The findings show that there is a weak positive dependence between return and volume in the tail. From an economic point of view, this implies that misunderstandings among investors may result in a reduction in liquidity in the cryptomarket and induce extreme price fluctuations. They also showed that the dependence structures of both Bitcoin and Ethereum are symmetric. In this context, Chemkha, BenSaïda, and Ghorbel (2021) also examined the dependence between cryptomarkets and currency markets using vine copulas. The studied currencies include the Japanese Yen (JPY), the euro, and the Great British Pound (GBP). Interestingly, their findings show that a weak dependence exists between cryptomarkets and conventionally studied currencies. This implies that global investors could diversify their portfolios by adding cryptocurrencies to obtain the benefits of having well-diversified portfolios. Maghyreh and Abdoh (2020) supported the previous findings by documenting weak tail dependence between Bitcoin and both the US dollar and euro.

Considering equity markets instead of currency markets, Borri (2019) demonstrated that cryptomarkets experience weak dependence on the US equity market. Lahiani, Jeribi, and Jlassi (2021) shed light on the dependence between cryptomarkets and equity markets belonging to BRICS developing countries and developed ones. They show that the introduction of Bitcoin futures changed the structure of dependence between cryptomarkets and equity markets. Jeribi and Fakhfekh (2021) investigated the tail dependence between cryptomarkets and US indices and documented modest tail dependence for all pairs of different assets. In other words, modest tail dependence is discovered between the US indices and each of the studied cryptocurrencies, namely, Bitcoin, Dash, Monero, Ripple, and Ethereum. Maghyreh and Abdoh (2020) documented the long-term right-tail dependence between Bitcoin and the S&P 500. Naeem and Karim (2021) performed an interesting piece of work by using the time-varying optimal copula (TVOC)

approach to examine the tail dependence between Bitcoin and green financial assets. Their findings show that the tail dependence between Bitcoin and green financial assets is characterized by multiple tail-dependence regimes and is also time-varying and asymmetric. This implies that green financial assets can be used by investors to hedge and provide more diversification benefits against Bitcoin. Cao and Xie (2021) discovered long memory and asymmetric multifractal features of tail dependence between cryptocurrencies and Chinese equity markets using the MF-ADCCA and time-delayed DCCA techniques.

With respect to the dependence between cryptomarkets and commodity markets, Borri (2019) and Lahiani, Jeribi, and Jlassi (2021) demonstrated that cryptomarkets experience weak dependence on the global gold market. Lahiani, Jeribi, and Jlassi (2021) demonstrated a unidirectional volatility spillover from Ethereum to the crude oil market and a unidirectional volatility spillover from the crude oil market to Bitcoin. Jeribi and Fakhfekh (2021) investigated the tail dependence between cryptomarkets and oil prices and documented modest tail dependence for all pairs of different assets. In other words, modest tail dependence was discovered between the oil price and each of the studied cryptocurrencies, namely, Bitcoin, Dash, Monero, Ripple, and Ethereum. Maghyereh and Abdoh (2020) demonstrated that the dependence between Bitcoin and commodities, silver, and oil, respectively, declined with their corresponding short-term medium return quantiles. Cao and Xie (2021) documented a modest long-memory feature of tail dependence between cryptocurrencies and Chinese gold after the COVID-19 shutdown. Yang and Hamori (2021) used the generalized autoregressive score-dynamic conditional score-Copula (GAS-DCS-Copula) technique to investigate the tail dependence between cryptocurrencies and the carbon market. They discovered asymmetric tail dependence between them, resulting in carbon, especially the European market for carbon, being able to successfully act as a hedging tool and a safe haven against cryptocurrencies.

The previous literature has left several gaps for future research. First, to the best of our knowledge, the tail dependence between cryptocurrencies and both Islamic financial assets and emerging market indices other than the Chinese one has not yet been investigated. Second, the tail dependence between each of the large and small market cryptocurrencies and conventional global currencies, as well as the US and other important equity market indices, deserves some attention from future researchers. Chemkha, BenSaïda, and Ghorbel (2021) suggested investigating tail dependence by employing the regular vine copula for the general structure of tail dependence.

Moreover, very little is known about the tail dependence of Bitcoin options, futures and conventional and unconventional assets to date.

3.3 Implied volatility

Implied volatility is a key concept in option pricing theory and can be defined as the market's prediction of future volatility that is derived from the option pricing models (Davidson, Kim, Ors, & Szakmary, 2001). Only a few papers in the literature addressed implied volatility due to the relative recentness of the introduction of cryptocurrency options. For instance, Rubbaniy, Khalid, and Samitas (2021) investigated the safe-haven features of the major three cryptocurrencies, namely, Bitcoin, Ethereum, and Ripple, during the COVID-19 pandemic. The wavelet coherence analysis is used to construct the global COVID-19 fear index. The findings indicate that the abovementioned cryptocurrencies can be beneficial to protect investors and hedge against risks during the COVID-19 pandemic. They also discovered that these three cryptocurrencies possess similar features to conventional assets. As a result, investors could not employ them to hedge against risks during the COVID-19 pandemic. In other words, they did not possess safe-haven features during the COVID-19 pandemic, when the implied volatility index was relied upon as a proxy for market volatility. Rubbaniy, Polyzos, Rizvi, and Tessema (2021) used daily data from January 2015 to June 2020 to examine the herding behavior toward the cryptocurrency market-specific implied volatility index (VCRIX) as a proxy for cryptocurrency-specific market fear. The paper reported many interesting findings. First, it found no evidence of herding in times of greater degrees of cryptocurrency specific market fear, meaning that investors regained their trust in asset pricing models during shutdowns. Furthermore, the paper documented herding behavior toward VCRIX during the COVID-19 pandemic, except for the shutdown period. The latter finding implied that investors can benefit from this finding not only to make reasonable investment decisions in times of high levels of market fear and uncertainty but also to grasp the impact of herding on their investment decisions.

Das, Roux, Jana, and Dutta (2020) studied the safe-haven features of Bitcoin against crude oil implied volatility (OVX). The findings indicate that Bitcoin is not a good hedging asset against OVX. As a result, investors may choose other assets to invest in over Bitcoin to hedge against OVX in different market states. Fang, Su, and Yin (2020) used the GARCH-MIDAS model to study the impact of news-based implied volatility (NVIX) as a proxy for investors' expectations of future uncertainty on the lengthy volatility of the major five cryptocurrencies, namely, Bitcoin,

Ethereum, Ripple, Litecoin, and the New Economy Movement. The findings show a significantly negative effect on the lengthy volatility of the abovementioned cryptocurrencies. This implies that investors' expectations possess more predictive power based on lengthy cryptocurrency volatility compared to economic fundamentals. Akyildirim, Corbet, Lucey, Sensoy, and Yarovaya (2020) investigated the relationship between cryptocurrency volatility and both US implied volatility (VIX) and European implied volatility (VSTOXX) as proxies for market fear. The findings show that cryptocurrencies not only increase with investor fear, but also that there is an interdependence between cryptocurrency volatility and financial market volatility. Furthermore, this interdependence grows substantially when the implied volatility of financial markets increases. Kim, Trimborn, and Härdle (2021) constructed an index for cryptocurrencies implied volatility (VCRIX), a "Fear Index", to capture the investor's expectation of future uncertainty in the cryptomarket in the absence of cryptocurrency derivatives for the vast majority of cryptocurrencies. Chi and Hao (2021) predicted the volatility of Bitcoin and Ethereum. To do so, they compared four GARCH-family models with implied volatility induced by an option pricing model to examine the accuracy of the prediction of market volatility for each. They found that GARCH-family models work better than option-implied volatility in forecasting future realized volatility.

Future researchers could try to work on the realized extreme volatility that derivative indices such as VIX can help us attain and which is expected to be induced by the behavioral component of option pricing (Kim, Trimborn, & Härdle, 2021). Moreover, institutional investors such as cryptocurrency exchange-traded funds (ETFs) have recently grown substantially in the cryptomarket. Another venue for future research could be investigating whether pricing efficiency improves over time after the introduction of derivative options and the growing existence of such investors (Chi & Hao, 2021). Furthermore, more research is needed on the accuracy of forecasting using implied volatility indices and the information content of these indices in the cryptomarket. In addition, to the best of our knowledge, little is known about the determinants of implied volatility in the cryptomarket. Finally, employing jump diffusion processes and continuous-time diffusion is of particular importance to measure the dynamics of indices of implied volatility over time.

3.4 Stochastic volatility

Stochastic volatility is the most commonly used method of modeling time-varying volatility in financial markets (Shephard & Andersen, 2009). It can be defined as the price movements or fluctuations of asset prices not being fixed, as assumed in the Black-Scholes option pricing model, but rather following time-dependent stochastic processes (Privault, 2021). Similar to implied volatility, a handful of research papers have studied stochastic volatility in cryptomarkets. Four groups of research streams are identified with respect to the stochastic volatility of cryptocurrencies. The first focused on the existence of stochastic volatility properties in the cryptomarket. For example, Phillip, Chan, and Peiris (2018) and Phillip, Chan, and Peiris (2019) examined the stylized facts of cryptocurrency volatility and provided evidence of the existence of many stylized facts, including stochastic volatility, that we should be aware of to evaluate the instability of cryptomarkets. Similarly, Chaim and Laurini (2018) and Kliber, Marszałek, Musiałkowska, and Świerczyńska (2019) used a stochastic volatility model with dynamic conditional correlation and confirmed that the standard linear GARCH is less flexible and does not suit the unique features and dynamics of Bitcoin. As a result, reliance on stochastic volatility models is essential. Dimpfl and Elshiaty (2021) provided evidence that Bitcoin contributes the most to the aggregate volatility of the cryptomarket.

The second group focused on the use of multivariate stochastic volatility models. Chaim and Laurini (2018) used multivariate stochastic volatility with discontinuous jumps to examine the volatility dynamics of the major nine cryptocurrencies. They discovered two periods of large volatility, namely in 2017 and early 2018. After that, the findings show a gradual reduction in volatility towards the end of the sample time span. Shi, Tiwari, Gozgor, and Lu (2020) extended the previous work by using a multivariate factor stochastic volatility model (MFSVM) to examine the stochastic volatility of the major six cryptocurrency prices, namely, Bitcoin, Ethereum, Ripple, Litecoin, Dash, and Stellar. The findings show that volatility increased substantially in the last few months of 2016 and mid-2017, while it leveled off from January 2018 onwards. It is also observed that there is a significant change in volatility from year to year.

The third focused on the inclusion of stochastic volatility to improve the quality of forecasting. Cross, Hou, and Trinh (2021) demonstrated that the inclusion of stochastic volatility and a heavy-tailed distribution improves the predictive power of volatility models and outperforms the random walk benchmark. Hafner (2020) added the deterministic and stochastic components to Spline-GARCH, which improves the performance of statistical tests of bubbles.

The fourth group focused on the relationship between stochastic volatility and other factors such as EPU and the day-of-the-week effect. Yen and Cheng (2021) extended the stochastic volatility model proposed by Wang and Yen (2019) to examine the impact of the EPU index on cryptocurrency volatility. Interestingly, fluctuations in the Chinese EPU are a good predictor of cryptocurrency volatility, while fluctuations in the US, Japanese, and Korean ones are not. Moreover, fluctuations in the Chinese EPU are significantly and negatively related to the future volatilities of Bitcoin and Litecoin. Therefore, both Bitcoin and Litecoin are effective hedging instruments against Chinese EPU risk. Ma and Tanizaki (2019) used a stochastic volatility model with time-varying variance to examine the effect of the day-of-the-week on Bitcoin volatility from January 2013 to December 2018. They found empirical evidence of large volatility on Thursday and Monday.

From the studies above, we notice that most papers used small numbers of cryptocurrencies, except Phillip, Chan, and Peiris (2018) and Phillip, Chan, and Peiris (2019). As a result, replicating the previous studies using a wider range of cryptocurrencies would be of interest to future researchers. Secondly, the use of high-frequency data to capture intraday dynamics is almost absent in the current literature on the stochastic volatility of cryptocurrencies. Hence, it is of particular importance for future researchers to take up this task. Thirdly, efficient techniques, such as the range-based stochastic volatility models proposed by Alizadeh, Brandt, and Diebold (2002), maximum likelihood estimation of stochastic volatility models proposed by Aït-Sahalia and Kimmel (2007), and dynamic factor multivariate stochastic volatility (DFMSV) proposed by Han (2006), deserve some attention from future researchers. Finally, researchers are recommended to merge regime-switching into stochastic volatility models, as introduced by Vo (2009). This enables us to model the stochastic volatility process when there is a shift in the mean.

3.5 Drivers of volatility

In this section, we explain the factors that may play a role in predicting cryptocurrency volatility. In this context, we distinguish between factors related to rational expectation theory and factors related to the behavioral theory of finance. Studies that adopted the former one suggested factors like adverse selection costs (Tiniç, Sensoy, Akyildirim, & Corbet, 2020), S&P 500 returns and VIX returns (L'opez-Cabarcos, P'erez-Pico, Piñeiro-Chousa, & 'Sevi', 2021), global geopolitical risk (GPR) (Aysan, Demir, Gozgor, & Lau, 2019), global real economic activity for bullish and bearish market states (Walther, Klein, & Bouri, 2019), leverage effect and EPU (Yu, 2019), news

coverage (Lucey, Vigne, Yarovaya, & Wang, 2022), day-of-the-week effect such as Monday and Thursday (Ma & Tanizaki, 2019), macroeconomic news (Corbet S. , Larkin, Lucey, Meegan, & Yarovaya, 2020), transaction activity (Gkillas, Tantoula, & Tzagarakis, 2021), oil market shocks (Yin, Nie, & Han, 2021), CBOE gold ETF volatility index (GVZ), VIX, Google trends, global geopolitical risk (GPR) and the global economic policy uncertainty index (GEPU) (Liang, Zhang, Li, & Ma, 2022), and the trade weighted USD currency index (Bystrom & Krygier, 2018) as good predictors of cryptocurrency volatility. Moreover, Karaömer (2022) shows that a negative correlation exists between the volatility of cryptocurrency policy uncertainty and the volatility of cryptocurrencies. In contrast, Smales (2022) documents a positive association between cryptocurrency policy uncertainty and volatility. Wang, Ma, Bouri, & Guo (2022) use the LASSO model to examine the drivers of volatility and find that macroeconomic factors are stronger than technical factors in predicting cryptocurrency volatility. Nguyen, Nguyen, Nguyen, Pham, & Nguyen (2022) document a negative relationship between both the Chinese interbank rate and the US federal funds rate, and stablecoin volatility. Salisu & Ogbonna (2022) show that incorporating the news effect improves the predictive power of the model during the COVID-19 pandemic relative to the historical average model.

Meanwhile, studies that adopted the behavioral theory suggested factors like investor sentiment (Güler, 2021; L'opez-Cabarcos, P'erez-Pico, Piñeiro-Chousa, & 'Sevi', 2021; Bukovina & Marticek, 2016), investor-driven internet search (Bystrom & Krygier, 2018), investor attention (Sabah, 2020; Al Guindy, 2021), Twitter-based sentiment (Kyriazis, Papadamou, Tzeremes, & Corbet, 2022), and herding behavior (Mandaci & Cagli, 2022) as good predictors of cryptocurrency volatility.

Many research gaps have been found in the current body of knowledge. For instance, examining the ability of monetary environment factors to predict cryptocurrency volatility is somewhat missing in the current literature. Moreover, other macroeconomic variables, such as gross domestic product (GDP), yield spread, inflation, changes in housing prices, US consumer confidence, business cycle factors, unemployment rate, and foreign direct investments, also remain unexplored. Researchers may also examine the ability of some institutional factors, such as corruption, transparency, and democracy, to predict cryptocurrency volatility. Secondly, future researchers have attempted to investigate the domestic political risk in huge economies, such as the USA and China, as potentially powerful predictors of cryptocurrency volatility, as suggested by Aysan, Demir, Gozgor, & Lau (2019). On the behavioral side, prior literature focused on only

three behavioral factors, namely investor sentiment, investor attention, and herding. Other behavioral biases, such as individualism, uncertainty avoidance, overconfidence, overreaction/underreaction, risk aversion, prospect theory, national culture, investor IQ, mental accounting, and representative heuristics, should be investigated separately to determine their contribution to the predictability of cryptocurrency volatility.

4. Discussion

Research on cryptocurrency volatility is still far from being a mature subject of study because knowledge about cryptocurrency volatility is just slightly improved by the current body of knowledge. Thus, a lot more research is needed to build well-established beliefs about this new financial world. This review has raised some impressions. First, advancing cryptocurrencies will greatly depend on developing more sensible theoretical explanations for how this new financial world works, resulting in a better understanding of the nature and key sources of cryptocurrency volatility in general and stylized facts, as discussed in Section 2.2, in particular.

Secondly, there is profound empirical evidence that the dynamics of cryptocurrency volatility are unique. Such dynamics are multifractality (Mensi, Lee, Al-Yahyaee, Sensoy, & Yoon, 2019; Kakinaka & Umeno, 2022). There is inverted asymmetric volatility compared to stock markets, and reactions to positive shocks outperform those to negative ones (Baur & Dimpfl, 2018; Cheikh, Zaied, & Chevallier, 2020; Fakhfekh & Jeribi, 2020); there is an unusually high level of uncertainty, and accordingly excessive volatility, compared to conventional assets due to the decentralized and deregulated nature as well as due to being unrelated to fundamentals or monetary policy (Klein, Thu, & Walther, 2018; Dyhrberg, 2016; Brauneis & Mestel, 2019); cryptocurrency markets are more prone to misbehavior and manipulations that leave significant impacts on prices compared to other markets (Katsiampa, Corbet, & Lucey, 2019); they show nonlinear behavior due to their complicated and chaotic nature and display high long-memory behavior, which gives the fractionally integrated GARCH (FIGARCH) model some superiority over other GARCH-family models (Mensi, Al-Yahyaee, & Kang, 2019; Fakhfekh & Jeribi, 2020); they show heteroskedastic price returns (Katsiampa, 2019b), have bubbles (Corbet, Meegan, Larkin, Lucey, & Yarovaya, 2018; Fry & Cheah, 2016), have extreme characterize cryptocurrency volatility because Bitcoin volatility is almost eight times greater than stock market volatility (Baur & Hoang, 2021; Harvey, 2017; Brauneis & Mestel, 2019; Ji, Bouri, Lau, & Roubaud, 2019); they have stochastic volatility (Phillip, Chan, & Peiris, 2018; Cross, Hou, & Trinh, 2021), are heavy tailed

(Phillip, Chan, & Peiris, 2018; Cross, Hou, & Trinh, 2021), involve regime change or structural breaks (Ardia, Bluteau, & Rüede, 2019), have mean reversion for minutely, hourly, daily, and weekly data frequencies (Corbet & Katsiampa, 2020; Turatti, Mendes, & Caldeira, 2020; Yarovaya, Matkovskyy, & Jalan, 2022), and contain volatility spillover (Katsiampa, 2019a; Okorie & Lin, 2020).

Thirdly, most papers in the literature used a version of the GARCH-family models. While these models offer some advantages, such as producing more accurate estimations of volatility and successfully dealing with the heteroskedasticity problem compared to the least squares methods, there is considerable evidence of the inappropriateness of this family of models to deal with the unique and unprecedented dynamics of cryptocurrency volatility. Also, GARCH-family models produce biased estimations where there are structural breaks. As a result, employing stochastic volatility models is duly justified. In other words, researchers are required to provide robust empirical volatility models and to conduct more replications and robustness research to support the currently available evidence, improve its validity and reliability, and stimulate its theoretical development. This may require solid and integrated econometric and financial knowledge as well as dedicated researchers who are willing to improve our understanding of this new financial world.

This review has important theoretical and practical implications for investors, researchers, regulatory bodies, portfolio managers, and policymakers because it impacts asset allocation, portfolio selection and management, risk management, and asset pricing, requiring them to be aware of the dynamics of cryptocurrency volatility. Meanwhile, being aware of the dynamics of cryptocurrency volatility will help cryptomarket participants make better investment decisions. Policymakers may use their understanding of these dynamics to set up new regulations that will improve the efficiency of financial markets and guide investors toward a better allocation of assets, leading to revived economic growth and development.

One prominent implication of this study is whether or not the current research indicates the market efficiency of cryptomarkets. There is prevailing evidence of inefficiency and an absence of random walk in cryptomarkets due to the existence of many stylized facts such as long memory, multifractality, structural breaks in means and variance, volatility clustering, persistence, asymmetry, and some behavioral biases like herding, among others. This inefficiency has influential effects on portfolio diversification, asset allocation, risk management, the valuation process, and the investment decision-making process. As a result, policymakers and regulatory bodies should work toward instituting the legal and regulatory infrastructure of cryptomarkets in

order to not only improve transparency and the informational environment but also reduce manipulation and misbehavior to eventually enhance efficiency levels. Recently, the growing attention of financial institutions to cryptomarkets may hopefully attract more rational investors to the market and relatively improve the level of market efficiency.

The second implication of this paper is the safe-haven and hedging capabilities of cryptocurrencies. Hedging capability means a specific asset is negatively related or perfectly unrelated to another asset, while a safe haven asset is an asset that is negatively related or perfectly unrelated to another asset in distressed markets only (Baur & Lucey, 2010). With respect to the hedging capabilities of cryptocurrencies, conflicting findings have been demonstrated in the literature. Some scholars doubt the capability of cryptocurrencies to act as good hedgers (Bouri, Molnár, Azzi, Roubaud, & Hagfors, 2017; Tiwari, Adewuyi, Albulescu, & Wohar, 2020; Klein, Thu, & Walther, 2018). Others argue that cryptocurrencies can be good short-term hedgers in extreme market states (Bouri, Gupta, Tiwari, & Roubaud, 2017; Demir, Gozgor, Lau, & Vigne, 2018). The third stream of research claims that Bitcoin has the same hedging capabilities as gold (Dyhrberg, 2016). Meshcheryakov and Ivanov (2020) provided evidence that Ethereum has hedging capabilities against the US stock and gold markets. Finally, Qiao, Zhu, and Hau (2020) found the long-term hedging capabilities of Bitcoin against other cryptocurrencies.

With respect to safe-haven capabilities, conflicting findings have also been found. The first group argues that Bitcoin does not have safe haven capabilities. Stablecoins such as Tether, USD coin, TrueUSD, Paxos, Dai, and Gemini Dollar can act as a safe haven against Bitcoin, Ethereum, Litecoin, and Ripple (Baur & Hoang, 2021; Mensi, Al-Yahyaee, Al-Jarrah, Vo, & Kang, 2020). Kliber, Marszałek, Musiałkowska, and Świerczyńska (2019) demonstrated that Bitcoin can be a safe haven against economic recession in Venezuela only among the studied countries. Rubbaniy, Khalid, and Samitas (2021) documented that Bitcoin, Ethereum, and Ripple have safe haven capabilities against risk during the COVID-19 pandemic. Yang and Hamori (2021) provided evidence that the European carbon market possesses safe-haven and hedging capabilities against cryptocurrencies.

In short, it is clear that there is no consensus on the hedging and safe haven capabilities of cryptocurrencies. Much research is needed to help tackle the conflicting and insufficient evidence in the literature and help us better understand the hedging and safe haven properties of cryptocurrencies.

5. Toward a new research agenda

Since providing a future research agenda is the key output of reviewing literature, we introduce some research gaps in the current body of knowledge that may help extending the cryptocurrency volatility theory in the literature. In this section, we suggest some future research directions, as follows: First, replicability, reproducibility, and robustness studies are of importance to cryptocurrency volatility in particular and to the FinTech subject of study in general. Although some scholars and leading journals underestimate such scientific studies because they do not advance knowledge from this point of view, we believe that replicability, reproducibility, and robustness are critical for theoretical development (Dau, Santangelo, & Witteloostuijn, 2021; Duncan, Engel, Claessens, & Dowsett, 2014). For instance, we suggest that future researchers replicate, reproduce, and conduct robustness studies by employing a wider range of cryptocurrencies to overcome the current weakness with respect to the use of a limited number of cryptocurrencies in most research to date. Other ways to replicate, reproduce, and conduct robustness studies could include the use of different data frequencies, different sample sizes, different econometric techniques, different volatility measures, subgroup analysis, different research designs, different time periods, and/or different market states. This way, researchers could enable practitioners and the academic community to develop deep-rooted thoughts and beliefs on the dynamics of cryptocurrency volatility and affirm the very presence of empirical results. Moreover, this would help overcome the limitations of individual research, which extends the available knowledge on cryptocurrency volatility and deepens the understanding of this financial phenomenon. In addition, these kinds of research serve to generalize of the original paper, make policymakers, investors, portfolio managers, and other market participants more certain about the potential impact of this decision, and inform them that their decisions are not based on biased findings.

Second, due to the lack of sufficient investigations and evidence, more focus on high-frequency data (i.e., hourly, minutely and secondly) is essential and should be included as part of the research agenda. This would enable us to understand the intraday dynamics and behaviors of cryptomarkets. The importance of high-frequency data increases the unique features of cryptomarkets., which operate 24 hours a day, 7 days a week, and are characterized by being fundamentally different, largely decentralized and totally unregulated with a lack of institutional investors. This may add some computational complexity but will assist in developing intraday

trading strategies. Furthermore, there is empirical evidence that the use of low-frequency data may lead to spurious results (Christensen, Oomen, & Podolskij, 2014).

Third, because of the relative shortage of machine learning models such as neural networks in cryptomarkets, we chose them as our third research direction. The digital revolution imposes two serious challenges to conventional statistical techniques: the unprecedented computational power and the massive amount of available data (Brooks, Hoepner, McMillan, Vivian, & Simen, 2019). This widens the use of neural network-based models for financial applications, including forecasting. These models have yet to spread in cryptomarkets. Neural network models provide some advantages over others, such as accounting for nonlinear and non-random dynamics in the forecasting process (Chen, Leung, & Daouk, 2003). The use of this bundle of forecasting models is expected to provide investors, portfolio managers, researchers, and other cryptomarket participants with accurate forecasts of the future price movements of cryptocurrencies, which leads to the development of profitable trading strategies. In the same vein, we noticed a lack of attention to prediction models of cryptocurrency's extreme volatility.

Fourth, the recent introduction of cryptocurrency derivatives in 2017 opens new space for a stream of research on cryptocurrency volatility. This includes all research related to implied volatility in the presence of derivatives rather than the absence thereof. This should include the information content of options' "implied volatility" on the expected volatility of the underlying assets. Moreover, the presence of Bitcoin options and futures may lead to an excess volatility phenomenon, which is the fluctuation in asset prices that does not belong to the price discovery. What induces this excess volatility, whether this is related to liquidity, and whether these derivatives would enhance stability and efficiency are interesting areas for research. Also, important issues such as, but not limited to, constructing implied volatility indices, the determinants of implied volatility, the relationship between implied volatility and return, and the relationship between implied volatility and realized volatility, should be considered.

Fifth, how investor behavior can explain the cryptocurrency volatility puzzle will play a critical role in understanding how this new financial world operates. Substantial work should be conducted to constitute a comprehensive picture of why and when crypto investors trade, what cryptos they select to include in their portfolios, how crypto investor behavior differs across different exchanges, what cognitive errors they make when trading, and finally how this behavior impacts crypto prices and volatility. In this regard, there is reasonable justification and evidence to believe that investors can help explain excessive volatility and that bubbles can play a role in

the performance of initial coin offerings, especially as rational expectations-based explanation theories have failed to provide sensible and robust explanations of the statistically sophisticated and nonlinear features of asset prices (Olsen, 1998). Even in cryptomarkets, there is initial evidence that investor behavior is more important than macroeconomic news in forecasting volatility (Fang, Su, & Yin, 2020). This is consistent with the view that not only the arrival of new information but also market states and group occasions such as herding can push asset prices to fluctuate (Shefrin, 2002).

Other behavioral biases committed by individual investors, such as overconfidence, disposition effect, loss aversion, mental accounting, ambiguity aversion, regret avoidance, conservatism, and framing, need to be considered in cryptomarkets., especially their role in explaining the stylized facts. Other behavioral phenomena, such as heterogeneous beliefs, investor myopia, the impact of different types of investors (i.e., individual investors, institutional investors, local investors, and foreign investors) on price fluctuations, social interaction, emotions, and optimism, may also play a role in inducing the stylized facts of cryptocurrency volatility.

With the new entry of institutional investors into the cryptomarkets, there is one more space for future research examining the impact of institutional trades on cryptocurrency prices and volatilities. In finance, the behavior of institutional investors is critical in causing price fluctuations and is sometimes more influential than fundamental news (Gabaix, Gopikrishnan, Plerou, & Stanley, 2006). Future researchers can investigate the impact of institutional trades on liquidity, cryptocurrency price informativeness and volatility, institutional herding, positive feedback trading, the institutional ownership of each cryptocurrency and its impact on price stability, and the national culture of institutional investors and its impact on style investing and price fluctuations.

Sixth, stablecoins (i.e., Tether) are a class of cryptocurrencies introduced in 2014 to provide investors with stable cryptocurrency prices and minimal price fluctuations (Hampl & Gyönyörövá, 2021). These stablecoins are usually backed and secured by other assets such as gold, USD, or a mixture of assets and are introduced to act as good hedgers against traditional cryptocurrencies (Wang, Ma, & Wu, 2020). Investigating the existence of stylized facts, drivers, appropriate estimating and forecasting econometric models, as well as the hedging and safe haven capabilities of these stablecoins, remains an unsolved issue, and we know very little about them. Future research can and should address these issues.

Seventh, despite the economic viability of conventional cryptocurrencies as an asset or as a store of value, they have been widely criticized for their harmful effect on the environment. For instance, a bitcoin transaction produces up to 1018 kg of CO₂ and requires 2143 kWh of electricity. This is equivalent to more than 73 days of electricity consumption for a U.S. household, while the CO₂ emission produced by one bitcoin transaction is more than what 2 million of VISA transactions produce (Ren & Lucey, 2022; Pham, Karim, Naeem, & Long, 2022; Arfaoui, Naeem, Boubaker, Mirza, & Karim, 2023). Chia, Cardano, Stellar, Tron, Ripple, Nano, Iota, Eos, Bitcoin Green, Algorand, Hedera, Polkadot, Flow, and Tezos are the prominent green cryptocurrencies. So far, very little is known about the price and volatility dynamics (i.e., stylized facts) of these cryptocurrencies. We highly recommend future researchers take care of this issue.

Eighth, similar to the green cryptocurrencies, the gold-backed cryptocurrencies as well as the gold-backed Islamic cryptocurrencies have lacked sufficient attention in the literature. Unlike most conventional cryptocurrencies, the value of gold-backed cryptocurrencies is linked to a tangible asset. As a result, their values are protected against unbearable price fluctuations. This feature is expected to uphold the safe-haven and hedging properties of this kind of cryptocurrency.

Finally, evaluating cryptomarket volatility forecasts is a promising area for future research. In general, volatility forecasts can be evaluated directly through some statistical methods that rank alternative volatility models. These evaluations can be conducted by using conventional techniques, such as those proposed by Mincer and Zarnowitz (1969) or Diebold and Mariano (1995), and their extensions in cases of realized variance, or by using the loss function when the available forecasts are extracted from more than one model. On the other hand, the indirect evaluation methods of forecasts rely on portfolio selection and value at risk (VaR) because volatility is involved in many economic situations, such as utility maximization, portfolio optimization, hedging and option pricing. Portfolio optimization techniques, dynamic portfolio management, and machine learning and their enormous extensions (either for estimation or modeling) provide much space for future research to assess the precise nature of volatility forecasts. Moreover, with respect to hedging risk, many scholars can depend on VaR and its robust extensions to assess volatility forecasts.

6. Conclusion

This paper provided a review and synthesis of isolated research points. We conclude that the dynamics of cryptocurrencies are unique and different from other financial and non-financial assets due to the unique foundations of new crypto exchanges and the distinctive features of these new financial products, such as being unauthorized and the presence of practical misconduct with respect to the use of cryptocurrencies. As a result, cryptocurrency volatility is worth extensive investigation.

We cannot claim that this review is without limitations. First, many relevant papers were neglected because they were published in poor-quality journals. In other words, we include papers only if they were published in a trustworthy academic journal. In other words, a paper was considered as long as its journal existed in the ABS journal ranking. Second, we preferred not to rely on some classifications of volatility that are based on the econometric models being used, such as pre-ARCH models, GARCH-family models, long-memory models, regime-switching models, and stochastic volatility models, to reduce the econometric color and to pay more attention to financial applications instead. Also, we cannot claim that this review provides a comprehensive picture of cryptocurrency volatility. It rather improves our understanding of this contemporary topic, which deserves more reviews and more classifications, especially with the huge number of available studies.

To sum up, the literature on cryptocurrency volatility has grown substantially over the few years. Nonetheless, there is still much room for more research to be conducted in this area. To be more precise, the use of high-frequency data, the use of a wider plethora of cryptocurrencies, and the use of advanced computational techniques such as neural networks and machine learning require some regard from future researchers. Moreover, the well-known stylized facts, such as asymmetric volatility, volatility spillover, structural breaks, volatility persistence, tail dependence, and extreme volatility, still have some avenues for future research.

Another important recommendation is whether stochastic volatility models work better than deterministic models in describing the unique dynamics of cryptocurrency volatility. This paper has reviewed some previous studies on this research point, but the door is still open for much more to be conducted. Finally, what determines future cryptocurrency volatility and the competition between conventional and behavioral variables in predicting it are unsolved issues with have fruitful theoretical and practical implications.

Declaration of competing interest

We declare that we have no conflict of interest.

Data availability

Data will be made available on request.

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